A Network Theory of Project Management*

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Abstract

Managers often have little discretion over pay, but they routinely decide who works on which projects. I study a manager who staffs projects by capping each worker's effort on each project. Workers prefer some projects to others and work harder when their teammates do, but find spreading effort across projects costly. Optimal assignments are characterized by a single ranking of projects, and two worker-specific cutoffs on it determine where each worker contributes freely, is held below his desired effort, or is kept off a project team. Because the ranking is common to all workers, project teams are nested. Standard network centrality can get the ordering of generalists and specialists backwards because the links that make a worker valuable to one team also make spreading him across several teams costly. Workers instead sort by a ratio of two centralities that captures their comparative advantage in spreading versus concentrating effort. Assignment selects the network through which shocks propagate, and an improvement at one project can degrade every other project in the organization.

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1 Introduction

How work should be divided among people is central to understanding how organizations function. Organizations typically manage multiple projects, which often benefit from team collaboration, and the division of labor across projects affects how hard each person works and how well projects are completed. While economic theory has largely focused on how organizations influence behavior through incentives (e.g. Holmström, 1979; Holmström and Milgrom, 1991), much less is known about how behavior can be shaped by assignments. Yet managers often have little day-to-day discretion over pay, and in settings where people are motivated by the work itself (Prendergast, 2007), what workers are assigned to is a margin that matters. Indeed, recent empirical research shows that how workers are assigned affects firm performance (Minni, 2026).[1] This paper studies how managers can optimally organize production by deciding who works on what, how much, and with whom.

In practice, assignment often takes the form of a cap. Under most National Institutes of Health career development awards, a physician cannot spend more than a quarter of his time with patients.[2] Firms in product development limit how many projects engineers can join at one time.[3] A consulting associate is staffed only part-time on a project he is passionate about and assigned to tasks for smaller clients instead. The famed Cavendish Laboratory instituted an evening curfew, limiting the time researchers could spend on experiments so that they would go home and think.[4] In each case, the organization values the restricted activity and the person is willing to contribute more effort, yet the limit is assigned rather than priced outright.

I model a manager who can cap each worker's effort on each project, and the caps a worker faces across all projects form his *assignment*. A cap can only limit how much a worker contributes and cannot directly compel effort. The manager values the profile of project qualities through an increasing, jointly concave function, where a project's quality is the total effort it receives. I focus on management within a business unit or

[1] See Benson and Shaw (2025) for a broader discussion of the literature on how managers create value through job allocation.

[2] NIH Career Development K awards stipulate a split of 75% full-time professional effort and 25% clinical work.

[3] Wheelwright and Clark (1992) argue that productivity drops as engineers take on more concurrent projects.

[4] Ernest Rutherford enforced the policy as director (Perutz, 1984), during a period in which the laboratory made groundbreaking discoveries in nuclear physics, including the neutron and the splitting of the atom. I thank Jin Li for bringing this example to my attention.

division (teams of analysts, engineers or consultants) where a unit of effort contributes the same to a project no matter who supplies it. Formally, the manager sets effort caps and workers exert effort at each project in a Nash equilibrium of the induced effort game.

What a manager can achieve depends on three economic forces that shape a worker's effort provision. First, the worker has intrinsic preferences that differ across projects (Kristof-Brown et al., 2005; Carpenter and Gong, 2016). Second, he values working alongside others. His marginal benefit of effort on a project rises with his teammates' effort there (Mas and Moretti, 2009; Chiocchio et al., 2011). Third, his effort is costly, and effort on one project raises the cost of effort on every other through multitasking (Leroy, 2009; Coviello et al., 2014). Framing the induced effort game as a network game featuring within-project complementarity from collaboration and cross-project substitutability from multitasking, I provide a spectral condition under which an equilibrium exists and is unique.

Each worker faces a tension reminiscent of the one between adaptation and coordination that is ubiquitous in organizations (Dessein and Santos, 2006). Since multitasking makes spreading effort across multiple projects costly, each worker must strike a balance between his preferences, which pull him toward his favorite projects, and complementarity, which pulls him toward projects his close collaborators are working on. By setting effort caps, the manager shifts that balance toward the projects she values most. To illustrate, return to the consulting associate whose manager deliberately limits how much he can work on his favorite project. The cap frees up his capacity for other projects, but his colleagues who enjoy working with him on that project withdraw their own effort there. Assignment allows the manager to mitigate the misalignment between her preferences and those of her workers, but she must do so while managing workloads and preserving the gains from collaboration.

To describe the first main result, I say that a worker is *assigned* to a project if he exerts positive effort there, and *capped* if his cap holds him strictly below the effort he would otherwise choose. Optimal assignments are organized around a single endogenous project ranking. When projects are ordered by decreasing marginal value of quality to the manager at the optimum, the assignment of each worker i is characterized by two project indices, $\phi_i \leq \alpha_i$, on that ranking. Worker i is assigned and left uncapped at the first ϕ_i projects, assigned but capped at projects $\phi_i + 1$ through α_i, and unassigned with a zero cap at all remaining projects (Theorem 1).

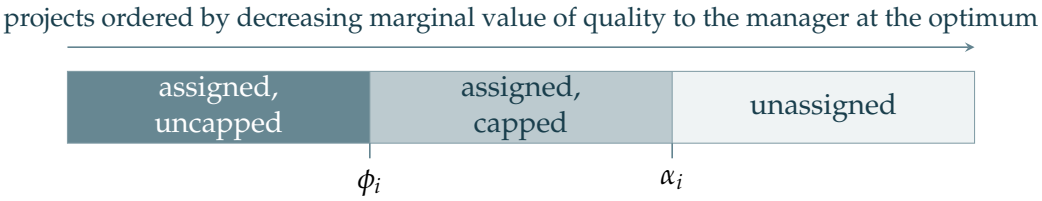


Figure 1: Optimal Worker Project Assignments

That optimal project assignments have an ordered structure is surprising given the rich heterogeneity across workers. One might expect the manager to exploit workers' comparative advantages and concentrate them on projects they prefer, forming distinct teams with strong complementarities that overlap irregularly, if at all. Theorem 1 rules this out, and its organizational implication is that project teams are nested. Each project's team contains the team of every lower-ranked project, so teams shrink as one moves down the ranking.

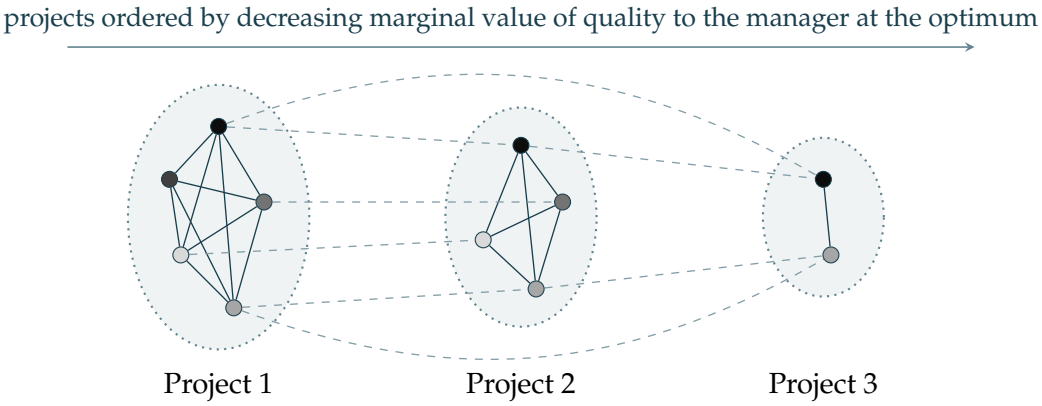


Figure 2: Nested Project Teams in the Optimum

Notes: Nodes are workers, and a node's shade identifies the same worker across projects. Solid edges join teammates at a project while dashed edges join the same worker across different projects.

The intuition for Theorem 1 comes from an arbitrage argument. Suppose a worker is assigned to a low-value project but capped or unassigned at a high-value one. Redirecting his effort toward the high-value project by lowering his cap at the former and raising it at the latter seems profitable. However, reducing his effort at the low-value project weakens complementarity for the teammates he leaves behind, and they will want to cut back on their contributions.

The solution is to coordinate the arbitrage across the *entire project team*: move a

small amount of every teammate's effort from the low-value project to the high-value one. Crucially, doing so offsets the complementarity the team loses at the former *and* exploits the extra complementarity it gains at the latter. Each teammate's shift changes the complementarity affecting others at both projects, triggering further adjustments. I show that a transfer pinned down by the *team interaction matrix* accounts for all such feedback simultaneously and preserves every worker's incentive constraint, so it is both feasible and profitable. Because the optimum admits no such arbitrage, no worker is ever capped at or unassigned from a high-value project while assigned to a low-value project.

The structural result reflects in part how multi-project organizations allocate workers. Practitioners often rank projects by financial value and retain only as many projects as available resources allow (Cooper et al., 1999), amounting to a single assignment cutoff on the ranking. Theorem 1 provides an optimality rationale for this heuristic and sharpens it in two ways. First, the project ranking is determined jointly with staffing in the assignment problem. Second, each worker has his own two cutoffs rather than sharing a single cutoff with the whole organization. The theorem also identifies optimal staffing as a novel source of nestedness, a regularity documented in many types of socio-economic networks (Mariani et al., 2019).

While the heuristic decides only which projects to staff, Theorem 1 also contains an intermediate regime in which a worker is assigned but capped. This regime requires the manager to choose exactly how much effort the worker exerts and monitor him to keep him from contributing more. I provide a necessary and sufficient condition for the optimal assignment to be coarse: every worker is either uncapped or unassigned at each project (Proposition 1). When the condition holds, deciding team membership alone is optimal and the manager acts only on the extensive margin of assignment, as the heuristic does. This is of practical interest because team membership is often easier to enforce than effort.[5] A worker assigned to and left uncapped at a project chooses his own effort there, and one unassigned from a project is simply kept off its team.

My second set of results concerns how labor is divided: which workers specialize only in the top-ranked projects (small ϕ_i and α_i), and which operate as generalists (large ϕ_i and α_i). In general, how workers sort cannot be read off the primitives. Whether a worker is assigned to a project, and whether he is capped there, depends

[5]Caps can still be enforced in practice. Management software such as Jira, Time Doctor and Clockify assists managers with monitoring and enforcing effort, while restricting access to necessary tools and equipment, as the Cavendish Laboratory did, limits effort directly.

on (1) who else is staffed there, which is determined by every other worker's cutoffs, and (2) how thinly he is already spread, which is dictated by his own cutoffs. Because costs and complementarity act through both channels at once, I disentangle the two forces by shutting down each channel in turn.

I first assume homogeneous complementarity, so the identity of a collaborator does not affect a worker's marginal benefit from collaboration. This shuts down the first channel; workers still differ in costs and preferences. Workers are then ordered by a *specialization index*, a closed-form statistic that increases in a worker's cost of multitasking and decreases in his cost of concentrating on a single project. A worker with a higher specialization index has weakly lower assignment and cap cutoffs, so specialists' assignments are nested within those of generalists (Theorem 2).

Next, I restore heterogeneous complementarity and assume every worker is assigned to every project. This shuts down the second channel, leaving only the question of where each worker is capped. A natural conjecture is that the most central workers, who are the best collaborators, should be left uncapped at the most projects. Letting them contribute freely everywhere would amplify effort throughout the organization. However, this conjecture can fail because doing so spreads a central worker thin, which weakens his contribution to each team. The links that make him central are exactly what make spreading him across multiple projects costly.

Workers are instead sorted by their *scope indices*, each the ratio of two Katz–Bonacich centralities computed on the same network of complementarities but with different cost weightings (Theorem 3).[6] The first centrality measures a worker's influence on all workers' effort *across all projects* and is high when multitasking is cheap for him, his collaborators, their collaborators, and so on. The second measures his influence *within a project* holding total efforts fixed and is high when multitasking is instead expensive along those same chains. Because both centralities aggregate the same collaboration links, a well-connected worker can score highly on both. The ratio, capturing a worker's comparative advantage in spreading effort across multiple projects rather than concentrating it, sorts workers. A worker with a higher scope index is left uncapped at a superset of the projects where a lower-index worker is. This sorting contrasts with single-activity network games, where a single Katz–Bonacich centrality governs equilibrium effort (Ballester et al., 2006).

[6] Without full assignment, the sorting statistic keeps the same ratio form but has no closed form. It is built from endogenous multipliers and recomputed on a shrinking network as workers are dropped from projects (Appendix D).

The managerial lesson here is that neither being well connected nor being skilled at multitasking is enough to qualify a worker to be a generalist. In fact, ranking by centrality can get the order backwards.[7] What matters is not how well connected a worker is but the composition of his links: whether he is surrounded by colleagues who multitask well or poorly, how effectively those colleagues' own peers multitask, and so on. Being a generalist is therefore as much a relational property as it is a personal trait, and a manager cannot identify her generalists without knowing how well different workers work together.

Finally, I derive comparative statics of the optimal assignment. With my parametric form, incentive constraints are linear in effort, so each one can be interpreted as a budget the manager faces: a worker's project preference is the income, and his complementarity and cost parameters act as prices. Under a regularity condition and holding the pattern of assignment and caps fixed, the manager's response to a shock admits a Slutsky decomposition (Theorem 4). The substitution effect, in which the manager redirects effort across projects, arises only when some assigned worker is capped at the optimum (Proposition 5).

When every assigned worker is uncapped, the income effect splits into two opposing components (Proposition 6). If a worker's preference for a project rises, he works more there, and the extra complementarity induces his teammates to work more there as well. This is the *within-project spillover*. The extra effort raises these workers' marginal costs, so they cut effort at other projects, leading to a loss in complementarity for their teammates. I call this second component the *network reallocation effect*. If every worker is assigned to every project, the network reallocation effect degrades the quality of all projects not subject to the initial shock. Under partial assignment, its impact on other projects depends on the exact pattern of assignment, and a shock to one project can even raise the quality of another (Proposition 7).

In the discussion, I demonstrate that effort caps implement effort profiles that no contract rewarding effort can (Proposition 8). Any profile in which a worker is capped at a project requires a contract that penalizes his additional effort there. Caps and wages push effort in opposite directions, so assignment is a distinct instrument rather than a substitute for pay. I also contrast caps with effort floors and examine how the results extend to environments in which workers face hard limits on total effort or effort complementarities are asymmetric.

[7] I show this in Example 2.

I see three main contributions. First, I study the formation of overlapping project teams through effort caps in a multitasking environment. Because teammates' efforts are complements within a project, how a worker splits his effort becomes a coordination problem for the manager across project teams. The solution nonetheless takes a simple form: a single ranking of projects with worker-specific cutoffs. The structural result rationalizes priority-based staffing and offers a new explanation for nestedness in organizations. Second, I show that Katz–Bonacich centrality, the standard measure that sorts equilibrium efforts in single-activity network games, can rank workers in the wrong order once effort is divided across project teams. What matters is a worker's comparative advantage in spreading effort rather than concentrating it. Third, I quantify how shocks propagate when the manager re-optimizes in response: a positive shock to one project can degrade the quality of others.

Related Literature

This paper contributes to three areas of research: network games, principal-agent theory and organization design.

Network games. Early theories of network games restrict players to choosing a single action with complementary or substitutable interactions between neighbors (Ballester et al., 2006; Bramoullé and Kranton, 2007; Bramoullé et al., 2014). Two recent papers, Claveria-Mayol et al. (2024) and Dasaratha et al. (2025), embed a network game within a moral-hazard problem; the latter also characterizes which agents are active under an optimal contract. In contrast, I study assignment through effort caps across multiple projects. I characterize project-specific active sets and distinguish where each worker contributes freely, is assigned but capped, or is excluded.

Work on network interventions posits a social planner who removes players or designs a vector of marginal benefits to influence activity (Belhaj and Deroïan, 2018; Galeotti et al., 2020, 2025). By determining how much complementarity workers transmit within a project, the manager in my model effectively selects which links within a network are active, and to what extent. Here, the network is designed, whereas in the network formation literature, link formation is decentralized (e.g. Jackson and Wolinsky, 1996; Bala and Goyal, 2000).

In multiplex network games, players choose multiple actions across several layers subject to an aggregate effort constraint (Chen et al., 2018; Zenou and Zhou, 2026).

Closest to this work is Sadler and Yeh (2026), in which players select neighbor-specific efforts subject to a convex cost of total effort; my effort game is a parametric multigraph version of it. The novelty here is that the manager selects the structure of the effort game, so the comparative statics are those of an endogenous network. In a separate project, I study modular task assignments where effort at a task is a strategic *substitute* for effort at locally related tasks (Yeh, 2026). Here, efforts within a project are strategic complements.

Principal-agent theory. Principal-agent models have focused on incentive design in multitask or multi-agent settings (Itoh, 1991; Holmström and Milgrom, 1991; Dewatripont et al., 2000; Che and Yoo, 2001; Corts, 2007). This paper combines both dimensions but replaces the contracting instrument with assignment.[8]

Assignment is closest to delegation, since setting an effort cap is akin to choosing a delegation set (Alonso and Matouschek, 2008). Delegation is traditionally driven by asymmetric information, and the principal balances granting an agent flexibility to use his private information against limiting undesirable actions resulting from agent bias. My setting has no private information, and I identify a distinct reason for restricting action sets. Because effort costs are convex, forbidding effort at one project allows the manager to obtain more effort at another. The delegation sets in my environment are also interdependent. Lowering a worker's effort cap at one project shifts his action set at every other project through multitasking costs and his teammates' action sets at that project through complementarity.

Organization design. In assignment and matching models with transferable utility, agents are typically sorted by a single scalar attribute (Becker, 1973; Kremer, 1993). With a collaboration network and multitasking costs, what sorts workers in my model is a ratio of two network centralities. Recent work by Freund (2025) shows that specialization within a firm generates coworker complementarity and concentration of talent within a small number of firms. In their model, workers belong to a single firm, and so the question of multi-team membership is moot.[9] My main structural result also connects to optimal hierarchies, which arise from limited managerial attention (Geanakoplos and Milgrom, 1991), limited decision-making capacity (Hart and

[8] Assignment problems are prominent in operations research (Koopmans and Beckmann, 1957). There, allocation is chosen subject to capacity constraints for a server, rather than workers' incentive constraints. See Gurvich and Van Mieghem (2015, 2018) for examples and Silva et al. (2021) for a recent survey.

[9] For additional work on the optimal composition of a team and how it is affected by size, diversity, and incentives, see Prat (2002); Georgiadis (2015); Fu et al. (2016); Glover and Kim (2021).

Moore, 2005), communication (Visser, 2000; Garicano, 2000) and grouping activities for synergy (Dessein et al., 2010). These are hierarchies of authority or information, while those in the nested project assignments are of team membership.

2 Model

There is one manager (she), a set of workers (he) $N = \{1, ..., n\}$ and a set of projects $P = \{1, ..., m\}$.

Workers. Worker i chooses nonnegative effort $e_i^p \in [0, \bar{e}_i^p]$ to exert at each project p where $\bar{e}_i^p > 0$. He receives utility given by the sum of per-project benefits less his total cost of effort.

$$U_i(\mathbf{e}_i, \mathbf{e}_{-i}) = \sum_{p=1}^{m} \underbrace{(\beta_i^p e_i^p + \sum_{j \neq i} g_{ij} e_i^p e_j^p)}_{\text{per-project benefit}} - \underbrace{\left(\sum_{p=1}^{m} \frac{c_i}{2} (e_i^p)^2 + r_i \sum_{p,q \,:\, p<q} e_i^p e_i^q \right)}_{\text{cost of effort}}$$

where $\mathbf{e}_i = \{e_i^p\}_{p=1}^m$ and $\mathbf{e}_{-i}$ represents other workers' efforts at all projects.

From project p, worker i receives a benefit that is linear in his effort. His intrinsic preference for working on project p is captured by $\beta_i^p > 0$. Effort complementarity between worker i and j is measured by $g_{ij} > 0$, which I assume to be symmetric i.e. $g_{ij} = g_{ji}$. Collaborative complementarity enters through worker motivation rather than production technology, aligning with recent work on team contracts with peer effects (Claveria-Mayol et al., 2024).

Worker i incurs a general quadratic cost of effort. His cost of specialization on a single project is $c_i > 0$, while his cost of multitasking is $r_i > 0$. To maintain convexity in costs, I assume that $c_i > r_i$.[10] Because the cost of effort is convex and benefit is linear, a worker's utility is concave in his own effort vector.

Manager. The quality of project p is given by the sum of worker efforts $Q^p := \sum_{i=1}^n e_i^p$. The manager values project qualities according to $V : \mathbb{R}^m \to \mathbb{R}$, which is strictly increasing in each argument, strictly concave and twice continuously differentiable.

[10] The Hessian of worker i's cost of effort is $(c_i - r_i)\mathbf{I}_m + r_i \mathbf{1}_m \mathbf{1}_m^\intercal$, which has eigenvalues $c_i + (m-1)r_i$ and $c_i - r_i$ with multiplicity of $m - 1$. For the cost to be strictly convex, its Hessian needs to be positive definite, which is true if $c_i - r_i > 0$.

The manager's choice of upper bounds $\bar{\mathbf{e}} \in \mathbb{R}_{+}^{mn}$ solves:

$$\max_{\bar{\mathbf{e}} \geq \mathbf{0}} \; V(Q^1, ..., Q^m)$$

s.t. $\mathbf{e}$ is the (manager-preferred) pure-strategy Nash equilibrium given $\bar{\mathbf{e}}$

If there are multiple equilibria induced by $\bar{\mathbf{e}}$, then the one that achieves the highest manager valuation is selected. In Section 4, I provide conditions under which there is always a unique equilibrium so no selection rule is required.

Remarks

1. There are no explicit contracts in this model. In Section 8.1, I consider an alternate model where the manager rewards effort. To implement outcomes in which workers are assigned but capped, the manager needs *negative* marginal rewards at those worker–project pairs.

2. Symmetric complementarities, together with a restriction on interaction strength introduced in Section 4, are sufficient for equilibrium uniqueness. Section 8.2 discusses how the assignment structure extends to asymmetric complementarities.

3. In the model, there is no physical upper limit on workers' efforts. The main structural result of optimal assignments remain robust to imposing an external hard limit on efforts (for example, a 40-hour work week with $\sum_p e_i^p \leq T_i$) as shown in Section 8.3.

3 An Illustrative Example

Before starting the main analysis, I first showcase the economics of assignments through an example with three workers $\{1, 2, 3\}$ and three projects $\{x, y, z\}$. I build up the model in three stages to isolate each force: preferences first, then multitasking costs, then collaboration. The manager's valuation is given by

$$V = 0.99 \ln(Q^x) + 0.87 \ln(Q^y) + 0.14 \ln(Q^z).$$

Worker i receives a payoff from project p that is linear in his effort e_i^p and pays a quadratic cost scaled by his specialization cost c_i. Recall that β_i^p captures intrinsic preferences and worker i's total payoff is $U_i := \sum_p U_i^p$.

$$U_i^p = \beta_i^p e_i^p - \frac{c_i}{2}\left(e_i^p\right)^2$$

	Preference $\beta_i^{(\cdot)}$			Cost
	x	y	z	c_i
Worker 1	0.46	0.96	0.91	0.70
Worker 2	0.50	0.99	0.87	0.67
Worker 3	0.57	0.65	0.95	0.79

Table 1: Model Primitives

The manager's instrument is an effort cap. She sets a ceiling $\bar{e}_i^p \geq 0$ for each worker-project pair (i, p) and the worker then chooses effort at each project to maximize his total utility subject to $e_i^p \leq \bar{e}_i^p$. At this stage, each worker's problem is additively separable across projects, so a cap at one project does not affect his effort provision anywhere else. Even though there is misalignment between the manager and her workers' preferences, it has no bite. It is optimal to set a cap at or above the effort level that solves a worker's first-order condition i.e. $\bar{e}_i^p \geq \beta_i^p / c_i$. The assignment problem is trivial, and multitasking costs and gains to collaboration are what make it interesting.

Multitasking. Workers incur multitasking costs from working on multiple projects.

$$U_i^p = \beta_i^p e_i^p - \frac{c_i}{2}\left(e_i^p\right)^2 - \underbrace{\frac{r_i}{2}\sum_{q \neq p} e_i^p e_i^q}_{\text{multitasking costs}}$$

	r_i
Worker 1	0.13
Worker 2	0.14
Worker 3	0.25

Table 2: Multitasking Costs

Multitasking costs make the structural misalignment in the preferences between the manager and workers a genuine problem. Panel A of Table 4 shows what workers do if they are left unconstrained i.e. $\bar{e}_i^p = \infty$. Each worker invests heavily in the project he cares most about (y or z), and exerts minimal effort at project x. This is suboptimal for the manager, who likes x the most.

Panel B of Table 4 gives the optimum. The manager uses caps to direct effort toward project x. She removes workers 2 and 3 from project z, which frees up capacity

for them to work at project x; their efforts nearly double there. Worker 1 is capped below what he would freely contribute, as his marginal benefit there strictly exceeds his marginal cost. Even though worker 3 prefers project z the most, he is taken off of it because his multitasking cost is much greater than those of others. The manager would rather he work on more valuable projects like x and y.

Collaboration. Workers enjoy working with teammates, as captured by the complementarity terms $g_{ij} = g_{ji}$. Due to multitasking costs, workers face a trade-off between working on projects they prefer and projects their collaborators focus on.

$$U_i^p = \beta_i^p e_i^p + \underbrace{\sum_{j \neq i} g_{ij} e_i^p e_j^p}_{\text{gains to collaboration}} - \frac{c_i}{2}\left(e_i^p\right)^2 - \frac{r_i}{2}\sum_{q \neq p} e_i^p e_i^q$$

	g_{ij}
Workers $(1,2)$	0.04
Workers $(1,3)$	0.13
Workers $(2,3)$	0.03

Table 3: Complementarities

Workers 1 and 3 are the strong pair. Panel C of Table 4 shows how collaboration changes the optimal staffing pattern. What matters is no longer just how well a worker multitasks, but who he multitasks alongside. One might expect the manager to place workers 1 and 3 at all three projects to exploit their outsized complementarities. In the optimum, she does the opposite: they are assigned to projects x and y and removed from z. Worker 2, who has the second-highest multitasking cost but collaborates weakly with both of them, covers z alone and ends up being the generalist.

	Multitasking Costs Only						With Collaboration		
	A. Unconstrained			B. Optimum			C. Optimum		
	x	y	z	x	y	z	x	y	z
Worker 1	0.25	1.13	1.04	0.28	1.16	**0.84**†	0.52	1.48	**0**
Worker 2	0.29	1.21	0.99	0.46	1.38	**0**	0.35	1.35	**0.81**†
Worker 3	0.28	0.42	0.98	0.51	0.66	**0**	0.61	**0.67**†	**0**
Quality $Q^{(\cdot)}$	0.82	2.77	3.01	1.25	3.20	0.84	1.48	3.50	0.81

† Marginal benefit strictly exceeds marginal cost.

Table 4: Worker Efforts and Project Quality

This example leaves two questions. Do optimal assignments take on a particular

structure, and which workers does the manager designate as specialists, and which as generalists? The main analysis answers both.

4 Implementability

To make the manager's problem tractable, I begin with characterizing its feasible set: the effort profiles that arise as an equilibrium of the induced effort game for some choice of caps. This takes three steps. I first ensure that every choice of caps induces a unique equilibrium (Lemma 1), then show that implementable profiles are exactly those satisfying worker incentive constraints wherever effort is positive (Lemma 2), and finally restrict parameters so that this set is convex (Lemma 3).

As is standard in network games, a spectral condition limiting the strength of network interactions guarantees that an equilibrium of the effort game induced the manager's choice of effort caps exists and is unique. To state it, I collect the cost and complementarity parameters into matrices.

Definition 1. The **specialization cost matrix** is $\mathbf{C} := \operatorname{diag}(c_i)_{i \in N}$, the **multitasking cost matrix** is $\mathbf{R} := \operatorname{diag}(r_i)_{i \in N}$, and the **complementarity matrix G** has off-diagonal entries $\mathbf{G}_{ij} = g_{ij}$ with zeros on the diagonal.

Assumption 1. *The spectral radius of* $(\mathbf{C} - \mathbf{R})^{-1}\mathbf{G}$ *is less than 1.*

The assumption does two things. First, as in Ballester et al. (2006), it bounds complementarity and prevents efforts from diverging to infinity. Second, and specific to this model, it bounds complementarity and multitasking costs jointly. A worker with high multitasking costs prefers to focus only on a few projects, and if complementarity is also high, then where she works depends on where her colleagues work. This creates a coordination problem and can produce multiple equilibria. Notice that subtracting $\mathbf{R}$ makes the condition tighter than without multitasking costs because $c_i > r_i$ implies each diagonal entry $1/(c_i - r_i)$ of $(\mathbf{C} - \mathbf{R})^{-1}$ exceeds $1/c_i$, so $(\mathbf{C} - \mathbf{R})^{-1}\mathbf{G} \geq \mathbf{C}^{-1}\mathbf{G}$ entrywise. The spectral radius of a nonnegative matrix is monotone in its entries.

Lemma 1. *Under Assumption 1, for every choice of effort caps, the induced effort game has a unique equilibrium.*

Proof. See Appendix A.1. □

Formally, the network game admits an exact potential because $\mathbf{G}$ is assumed to be symmetric (Monderer and Shapley, 1996). Assumption 1 ensures the exact potential is strictly concave. Because the effort space is compact, the exact potential attains a unique maximizer corresponding to the unique equilibrium.

4.1 Incentive Constraints

To simplify the manager's problem, I show that it is without loss of generality for the manager to directly choose efforts so long as each worker's *incentive constraint* holds whenever effort is positive.

Lemma 2. *For* $\mathbf{e} \geq \mathbf{0}$*, there exists an upper bound* $\bar{\mathbf{e}}$ *that implements* $\mathbf{e}$ *as an equilibrium if and only if*

$$\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q$$

whenever $e_i^p > 0$*. Moreover, the manager can implement* $\mathbf{e}$ *by setting* $\bar{\mathbf{e}} = \mathbf{e}$*.*

Proof. See Appendix A.2. □

When a worker's marginal utility from working on a project is nonnegative, reducing effort is never profitable. Capping effort at the manager's target level prevents the worker from contributing more even if he would like to. The listed inequality is worker i's **incentive constraint** at project p. The left term is worker i's marginal benefit at project p. It depends on his intrinsic preference and the efforts of other workers at that project. An increase in other workers' efforts at project p relaxes his incentive constraint. The right term is worker i's marginal cost at project p, which is determined by the efforts he supplies at other projects. Raising his effort cap at project p will not encourage more effort if his incentive constraint binds. Instead, the manager has to lower his effort caps on other projects or increase the efforts of his collaborators on project p.

4.2 Set-up: A Convex Optimization Problem

Implementing zero effort at a project requires no incentive constraint. It is important to disentangle the reasons why a worker exerts zero effort in equilibrium: (1) despite his willingness to contribute, he is constrained by an effort cap of zero or (2) he is not willing to contribute there due to a negative marginal utility at zero effort. In the second scenario, the worker is strictly worse off when starting work on a project, which falls outside the scope of this paper. To reiterate, I analyze when the manager restricts activity that both she and a worker value. Ruling out the second scenario also keeps the analysis tractable; otherwise, we need to track whether an incentive constraint needs to be satisfied or can be safely ignored at zero effort. The manager's problem becomes combinatorially difficult due to the potential nonconvexity of the feasible set.[11]

Accordingly, I restrict parameters so that workers are always motivated to start work on a project, and are forced off a project only if the manager sets their effort cap to zero. That way, the set of implementable effort profiles coincides with the set of profiles satisfying worker-project incentive constraint, including those at zero effort. What must hold is worker i's incentive constraint at zero effort on project p:

$$\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq r_i \sum_{q \neq p} e_i^q.$$

To satisfy it, his multitasking costs r_i cannot be too high relative to his intrinsic preference and his total effort. The latter is endogenous, and I bound it by summing his incentive constraints across projects and collecting terms in total effort, which I denote as $E_i := \sum_{p=1}^m e_i^p$

$$\sum_{p=1}^m \beta_i^p + \sum_{j \neq i} g_{ij} E_j \geq (c_i + (m-1) r_i) E_i$$

Regardless of where a worker allocates it, raising his total effort incurs a unit increase in specialization cost and a $(m-1)$-unit increase in multitasking costs. A rise in worker j's total effort reduces worker i's marginal cost by g_{ij}. Stacking across all workers gives a system in total efforts governed by the *aggregate interaction matrix.*

Definition 2. The **aggregate interaction matrix** is $\mathbf{P} := \mathbf{C} + (m-1)\mathbf{R} - \mathbf{G}$.

[11] Appendix B formulates the manager's problem as a mixed-integer nonlinear program and gives an explicit example of a nonconvex feasible set.

The entries of $\mathbf{P}$ map changes in *total* effort to changes in a worker's marginal cost of *total* effort. The stacked inequalities read $\mathbf{PE} \leq \sum_{p=1}^{m} \boldsymbol{\beta}^p$, so any effort profile satisfying every incentive constraint obeys $\mathbf{E} \leq \tilde{\mathbf{E}} := \mathbf{P}^{-1} \sum_{p=1}^{m} \boldsymbol{\beta}^p$.[12]

Assumption 2. *For each worker i, $r_i \tilde{E}_i < \min_p \beta_i^p$.*

A worker's total multitasking burden never outweighs his preference for his least favorite project, guaranteeing he is always strictly motivated to start work at any project. When preferences are uniform i.e. $\beta_i^p \equiv \beta$ for all (i, p), a simple sufficient condition for Assumption 2 in the primitives is $c_i - r_i \geq r_i(\sum_{j \neq i} g_{ij}/r_j)$. While the model already assumes $c_i - r_i > 0$, this condition asks for more. The cost of specialization must be high and that of multitasking low enough to dominate the pull from collaborators who are good at multitasking. The assumption rules out parameters under which a worker would rather work a lot on one project alongside his strongest collaborators than start on a new one. Most importantly, the assumption permits worker heterogeneity: they may differ in how well they multitask, who collaborates better with whom, and which projects they like most or least. These differences remain subject to the joint bound above. With this, the manager solves a tractable convex optimization problem.

Lemma 3. *Under Assumptions 1 and 2, every implementable effort profile satisfies $\mathbf{E} \leq \tilde{\mathbf{E}}$, implying the set of implementable effort profiles coincides with the set of effort profiles that satisfy incentive constraints for all worker-project pairs. The manager's problem is:*

$$\begin{aligned}
&\max_{\mathbf{e} \geq \mathbf{0}} \; V(Q^1, ..., Q^m) \\
&s.t. \; \forall (i,p), \; \beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q \\
&\forall p, \; \sum_{i=1}^{n} e_i^p = Q^p
\end{aligned}$$

Proof. See Appendix A.3. □

A solution to the manager's problem exists because the objective is continuous and the feasible set is closed and bounded. Moreover, the solution is unique in project qualities because the objective is strictly concave over total efforts and the feasible set is convex.

[12] I show in Appendix A.3 that $\mathbf{P}^{-1}$ exists and is nonnegative.

5 Structure of Optimal Project Assignments

Returning to the first question raised by the illustrative example: do optimal assignments take a particular structure? To describe staffing patterns, I classify each worker-project pair by two properties.

Definition 3. In effort profile $\mathbf{e}$, worker i is **assigned** to project p if $e_i^p > 0$, otherwise he is **unassigned**. Conditional on being assigned, worker i is **uncapped** at project p if his incentive constraint binds i.e. $\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p = c_i e_i^p + r_i \sum_{q \neq p} e_i^q$. He is **capped** if $\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p > c_i e_i^p + r_i \sum_{q \neq p} e_i^q$. Let $A^p := \{i \mid e_i^p > 0\}$ and $\Phi^p := \{i \mid i \text{ is uncapped at } p\}$ denote the set of workers assigned or uncapped on project p, respectively.

A worker is uncapped if he contributes his desired effort level up until his marginal benefit meets his marginal cost. This can be implemented by any effort cap that weakly exceeds that effort level. Being capped prevents a worker from contributing as much as he wishes. While unassigning worker i from project p eliminates p as a source of multitasking costs for other projects, it forfeits the complementarity he generates for other workers at project p. By assigning but capping him, the manager leverages him as a source of complementarity for others on a particular project while preserving his capacity to work on other projects.

The main structural result demonstrates that both assignment and caps are ordered by a common endogenous ranking. I write $V^p := \partial V / \partial Q^p$.

Theorem 1. *Assume that the strict complementarity at the optimum is satisfied. Rank projects by the manager's marginal value in the optimum so that $V^1 \geq \cdots \geq V^m$. For each worker i, there exist natural numbers $\phi_i, \alpha_i \in \{1, ..., m\}$ where $\phi_i \leq \alpha_i$ such that he is*

- *uncapped at projects $1, ..., \phi_i$,*
- *assigned but capped at projects $\phi_i + 1, ..., \alpha_i$,*
- *and not assigned to projects $\alpha_i + 1, ..., m$.*

This implies

$$\Phi^1 \supseteq \cdots \supseteq \Phi^m \quad \text{and} \quad A^1 \supseteq \cdots \supseteq A^m.$$

Proof. See Appendix A.4. □

I show in Appendix C that strict complementarity is generic in the optimum. For each worker, two cutoffs determine which projects he is assigned to and uncapped at. His project assignments comprise three successive regimes. For the most valuable projects, the manager allows him to work as much as he is willing to. For medium-value projects, the manager assigns but caps him. The worker would like to work more, but is prohibited from doing so. On the remaining low-value projects, he remains unassigned. All worker assignments agree on the *same* ranking of projects, namely by the manager's marginal value in the optimum. Project teams are necessarily nested as we saw in Figure 2.

That optimal project assignments embody such a clean structure is surprising, especially because all worker attributes— preferences, synergies, and costs— are heterogeneous. One may expect workers to retain a comparative advantage at certain projects over others, and it seems natural that workers specialize in distinct projects and have non-overlapping project assignments. Theorem 1 precludes any worker from being capped or excluded from a project while being uncapped or assigned to a less valuable one.

Why then should worker assignments be ordered this way? There are two pieces to the puzzle. Between projects the manager values differently on the margin, the ordering follows from a transfer of efforts that would otherwise be profitable (Section 5.1). Suppose a worker is assigned to a low-value project but would like to work more at a high-value one. Moving his efforts alone is infeasible because it reduces complementarity at the project team he leaves behind. Instead, I design a *team-coordinated* effort transfer from the low-value project to the high-value one that is both feasible and profitable. For projects with the same marginal valuation, I show that they have the same sets of workers who are assigned and uncapped (Section 5.2). Each worker's assigned-but-capped projects share a common marginal valuation and therefore lie on a marginal-value "plateau." Two additional results refine Theorem 1. First, I provide conditions under which the optimal assignment is coarse: every worker is either uncapped or unassigned at each project (Section 5.3). Second, although an endogenous ordering is a consequence of the model's generality, I give conditions under which the project order is exogenously determined (Section 5.4).

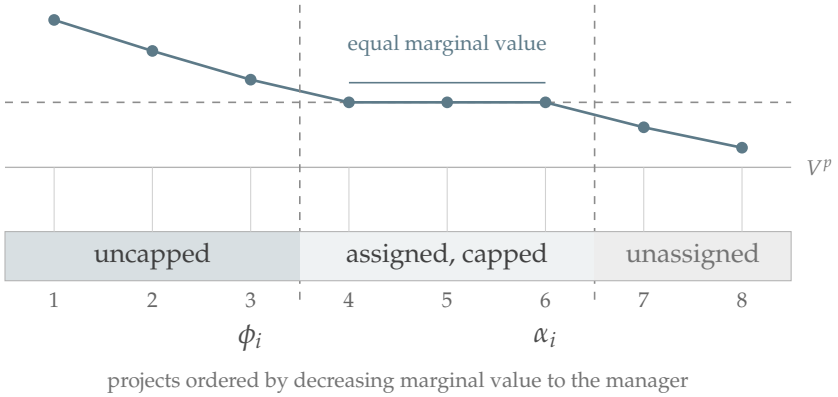


Figure 3: Optimal Worker Project Assignment

5.1 A Team-Coordinated Effort Transfer

To ensure that an effort transfer for workers between projects with different marginal values is feasible, I track the **slack** (s_i^p) of an incentive constraint, which must always remain nonnegative. Substituting total effort $\sum_{q \neq p} e_i^q = E_i - e_i^p$ separates how effort at project p and total effort affect slack there.

$$s_i^p = \beta_i^p + \sum_{j \neq i} g_{ij} e_j^p - (c_i - r_i) e_i^p - r_i E_i.$$

Suppose there are two projects p and q with $V^p > V^q$ in the optimum and worker i is assigned to q, but capped or unassigned at p, so he has positive slack at the latter. Consider the naïve transfer that moves $\delta > 0$ of his effort from q to p, which is profitable for small enough δ because

$$\Delta V = (V^p - V^q)\delta + O(\delta^2).$$

This transfer keeps worker i's total effort constant, so it leaves his slack at every other project unchanged. At project p, his slack falls by $(c_i - r_i)\delta$, which is absorbed by his existing slack. Every other worker's slack at p increases from the extra complementarity. The feasibility problem arises at project q. Reducing worker i's effort decreases complementarity at q for his teammates there, cutting their slack and potentially violating their incentive constraints.

The key insight is to coordinate the effort transfer on the entire project team at q rather than just worker i. For simplicity, I assume for now that every worker is

assigned to project q i.e. $A^q = N$; I return to the case when $A^q \subset N$ later. I design a transfer $\Delta\mathbf{e} \in \mathbb{R}^n$ that moves worker j's effort from q to p by Δe_j satisfying two requirements: (1) worker i's slack at p decreases by $\delta > 0$ and (2) every other worker's slack at both projects remains unchanged. Because the transfer preserves total effort, the change in slack is given by

$$\Delta s_j^p = \sum_{k \neq j} g_{jk} \Delta e_k^p - (c_j - r_j)\Delta e_j^p = -\Delta s_j^q.$$

The two requirements put together give $\mathbf{M}\Delta\mathbf{e} = \delta\boldsymbol{\epsilon}_i$ where $\mathbf{M} := \mathbf{C} - \mathbf{R} - \mathbf{G}$ and ϵ_i is the i-th basis vector.[13] Rather than invert $\mathbf{M}$ directly, I build intuition for what this transfer scheme achieves by leveraging a "round-by-round" accounting of how each worker's slacks change every time effort for another worker is transferred.

Round 1: The manager only transfers worker i's effort by $\Delta e_i^{(1)} = \delta/(c_i - r_i)$. This delivers the $-\delta$ change in his slack at p. Notice that for every other worker $j \neq i$, his slack at p expands by $g_{ji}x_i^{(1)}$ while his slack at q contracts by the same amount.

Round 2: The manager transfers efforts for every other worker $j \neq i$ to absorb his extra slack at p and erase his deficit at q, given by

$$\Delta e_j^{(2)} = \frac{g_{ji}}{c_j - r_j} x_i^{(1)}$$

This transfer creates additional slack at p and deficit at q for j's collaborators, which calls for a third round of transfers, and so on. The vector of transfers for each round satisfies a recurrence relation $\Delta\mathbf{e}^{(t)} = (\mathbf{C} - \mathbf{R})^{-1}\mathbf{G}\Delta\mathbf{e}^{(t-1)}$. That is, effort in round t travels along walks of length $t-1$ within the network of complementarities, weighted by complementarity g_{ji} and discounted by each worker's net cost $c_j - r_j$ of shifting effort between projects holding total efforts fixed. Summing over all adjustments (or walks) gives

$$\Delta\mathbf{e} = \sum_{t=1}^{\infty} ((\mathbf{C} - \mathbf{R})^{-1}\mathbf{G})^{t-1}(\mathbf{C} - \mathbf{R})^{-1}\delta\boldsymbol{\epsilon}_i$$

Assumption 1 prevents complementarity from causing the adjustment process to explode and rules out divergence. To state the formal result, I now give a name to $\mathbf{M}$, whose inverse is the limit of convergence.

Definition 4. The **team interaction matrix** is $\mathbf{M} := \mathbf{C} - \mathbf{R} - \mathbf{G}$.

[13] Recall $\mathbf{C} := \text{diag}(c_i)_{i \in N}$, $\mathbf{R} := \text{diag}(r_i)_{i \in N}$, and $\mathbf{G}$ has entries $\mathbf{G}_{ij} = g_{ij}$ and 0 on its diagonal.

The $\mathbf{M}$ matrix governs how worker interactions affect workers' marginal cost of effort at a project, *holding total efforts fixed.* The off-diagonals, which are $-g_{ij}$, capture complementarity between workers. The diagonal term $c_i - r_i$ measures the change in worker i's marginal cost of effort at project p while keeping his total effort constant. An extra unit of effort at project p costs c_i, and to keep total effort constant, he must give up a unit of effort at another project q for a refund of r_i. This shift in effort *raises* his marginal cost at project p by $c_i - r_i$. The next result establishes that the series of effort adjustments converges to $\mathbf{M}^{-1}$.

Lemma 4. *Under Assumption 1,* $\mathbf{M}$ *has a nonnegative inverse that satisfies*

$$\mathbf{M}^{-1} = \sum_{t=1}^{\infty} ((\mathbf{C} - \mathbf{R})^{-1}\mathbf{G})^{t-1}(\mathbf{C} - \mathbf{R})^{-1}.$$

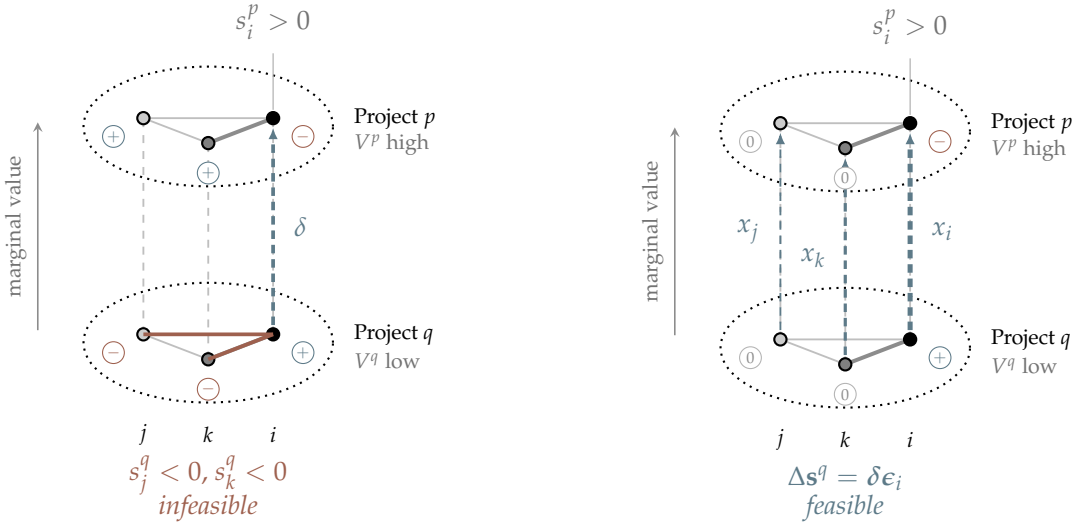
Proof. See Appendix A.5. □

The effort transfer has a simple formula: $\Delta\mathbf{e} = \mathbf{M}^{-1}\delta\boldsymbol{\epsilon}_i$. Qualitatively, each worker's transfer reflects his direct and indirect connections to worker i, weighted by the net costs of shifting effort across projects. It is profitable for sufficiently small δ because

$$\Delta V = (V^p - V^q)\mathbf{1}_n^\intercal \Delta\mathbf{e} + O(||\mathbf{x}||^2)$$

In the general case where $A^q \subset N$, the matrix of interest is $\mathbf{M}^{(A^q)}$, which is the submatrix of $\mathbf{M}$ with rows and columns restricted to only the workers on project team q. It inherits key properties of $\mathbf{M}$, including positive definiteness and a nonnegative inverse, and so a similar argument succeeds. I leave the technical details to the main proof in Appendix A.4.

Stated plainly, assigning a worker to a project with low marginal value but holding him back on one with high marginal value presents an arbitrage. The strategy is to redistribute the entire project team's effort towards the high-value project while respecting team members' incentive constraints. The transfer argument clarifies why heterogeneous project preferences do not generate disjoint specialization. Preferences enter incentive constraints additively and determine initial slack but disappear from its change under a transfer. Because cost and complementarity coefficients are project-invariant, the same team interaction matrix $\mathbf{M}$ governs slack changes at both projects when worker totals are held fixed. The manager can therefore coordinate transfers to preserve teammates' incentive constraints. Finally, because effort contributes equally



(a) Naïve transfer $\Delta \mathbf{e} = \delta \boldsymbol{\epsilon}_i$.

(b) Team transfer $\Delta \mathbf{e} = \mathbf{M}^{-1} \delta \boldsymbol{\epsilon}_i$.

Figure 4: Worker i and team-coordinated transfers where symbols $(+)$, (0) and $(-)$ indicate whether slack increases, stays the same or decreases.

to quality regardless of who supplies it, the coordinated transfer is profitable whenever the receiving project has a higher marginal value. Optimality rules out such staffing patterns.

5.2 Projects with Equivalent Marginal Values

Projects with the same marginal values in the optimum obey nested staffing patterns; in fact, these projects necessarily have identical sets of workers who are assigned and uncapped.

Lemma 5. *Projects with equal marginal value in the optimum have identical sets A^p and Φ^p sets. Moreover, each worker's capped projects share a common marginal value $V^{\phi_i+1} = \cdots = V^{\alpha_i}$.*

Proof. See Appendix A.6. □

Two projects with the same marginal value pose the same staffing problem to the manager. Positive definiteness of the team interaction matrix ensures the marginal cost of effort accrued by a worker outweighs the collaboration bonus he receives. This guarantees a unique solution to the staffing problem by ruling out coordination failures that lead to multiple assignment/cap solutions. The second statement can be

directly shown using a transfer argument. Suppose the manager caps a worker at two projects, p and q, in the optimum. She can perform the same team-coordinated effort transfer as before if there is an imbalance in marginal values. For example, if $V^p > V^q$, then she can pull efforts from q towards p for a profitable deviation. Therefore, it must be that both projects share the same marginal value.

It is tempting to read Lemma 5 as making the interval of projects with caps fragile because they must have equivalent marginal values. This is inaccurate: the effort for an capped worker-project pair is defined by two *strict* inequalities on the primitives—a slack incentive constraint and a slack nonnegativity constraint. The example in Section 3 has a unique optimum and so the cap pattern there survives small perturbations in every parameter.

5.3 When to Cap and Unassign

When a cap binds, the worker would like to contribute more effort than the cap allows. The manager must therefore choose the exact level of effort she wants and then extensively monitor the worker to keep him from exceeding it. Two of the three regimes of Theorem 1 avoid this problem: an uncapped worker chooses his own effort, so there is no enforcement needed and an unassigned worker never joins the project. The manager implements the latter at the extensive margin: she does not onboard him or give him access to the team, its tools, or the knowledge he would need to contribute. Whether a worker is on a team is easy to verify, but how much effort he exerts is harder to verify. I now provide a necessary and sufficient condition under which every worker is either uncapped or unassigned on every project, so that no cap needs to be enforced.

Definition 5. The worker interaction matrix is $\mathbf{W} := \mathbf{C} \otimes \mathbf{I}_m + \mathbf{R} \otimes (\mathbf{1}_m \mathbf{1}_m^\intercal - \mathbf{I}_m) - (\mathbf{G} \otimes \mathbf{I}_m)$ where $\otimes$ is the Kronecker product.

Proposition 1. *For a set of worker-project pairs A, let $\mathbf{e}(A) \in \mathbb{R}^{mn}$ be the effort profile with $\mathbf{e}_A(A) := \mathbf{W}_{(A)}^{-1} \boldsymbol{\beta}_A$ and $\mathbf{e}_{A^c}(A) := \mathbf{0}$, where $\mathbf{W}_{(A)}$ is the principal submatrix of $\mathbf{W}$ on A and $\mathbf{W}_{A^c A}$ is the submatrix with rows in A^c and columns in A. In the optimum, every worker is either uncapped or unassigned at each project if and only if there exists some A such that*

1. $\mathbf{e}_A(A) > \mathbf{0}$,
2. $\mathbf{W}_{(A)}^{-1} \nabla_{\mathbf{e}_A} V(\mathbf{e}(A)) > \mathbf{0}$, *and*

3. $\mathbf{W}_{A^cA}\,\mathbf{W}_{(A)}^{-1}\nabla_{\mathbf{e}_A}V(\mathbf{e}(A)) - \nabla_{\mathbf{e}_{A^c}}V(\mathbf{e}(A)) > \mathbf{0}$.

Proof. See Appendix A.7. □

Each row of the worker interaction matrix $\mathbf{W}$ corresponds to an incentive constraint. Conditions 2 and 3 are the Karush–Kuhn–Tucker conditions at the profile in condition 1, so together the three conditions are necessary and sufficient. Condition 1 requires that every worker, when uncapped, contributes positive effort at each assigned project. Condition 2 requires that the principal values more of every assigned worker's intrinsic motivation, net of the spillovers it generates through complementarity and multitasking, so no worker requires a cap. Condition 3 ensures that no unassigned worker is worth adding to a project team. The manager's marginal value of his effort at the project does not cover the multitasking burden he would impose on his other assignments, net of the complementarity he would generate. When such a set of worker-project indices A exists, caps only determine the extensive margin of assignment, not the intensive margin. Conversely, some worker is capped precisely when no team structure satisfies all three conditions.

The next result concerns when the manager needs to unassign or cap workers at all. As a benchmark, suppose the manager keeps every worker uncapped at every project at all: $\bar{e}_i^p = \infty$ for all (i, p). Stacking all incentive constraints gives $\mathbf{We} \leq \boldsymbol{\beta}$. Workers exert effort in the unconstrained equilibrium $\hat{\mathbf{e}} := \mathbf{W}^{-1}\boldsymbol{\beta}$, which I call the **no-intervention outcome.** Every incentive constraint binds in $\hat{\mathbf{e}}$, which is guaranteed to be strictly positive by Assumption 2.

Lemma 6. *Under Assumption 2,* $\hat{\mathbf{e}} > \mathbf{0}$.

Proof. See Appendix A.8. □

The no-intervention outcome is the unique assignment that assigns and keeps every worker uncapped at every project. As we saw in the example of Section 3, effort caps mitigate the misalignment between the manager's valuation and the workers' equilibrium contributions, with the latter shaped not only by workers' intrinsic preferences, but also by the primitives governing collaboration and effort costs. I provide a diagnostic of when this misalignment occurs.

Proposition 2. *The no-intervention outcome is optimal if and only if* $(\mathbf{W})^{-1}\nabla_{\mathbf{e}}V(\hat{\mathbf{e}}) \geq \mathbf{0}$.

Proof. See Appendix A.9. □

The diagnostic is a computation in the primitives. If $(\mathbf{W})^{-1}\nabla_{\mathbf{e}} V(\hat{\mathbf{e}}) \geq \mathbf{0}$, then it is optimal to simply let the workers choose efforts as they wish. If the vector yields a negative entry, then the manager must unassign or cap in the optimum.

5.4 Exogenous Project Orderings and Cutoffs

The ordering of projects in Theorem 1 is endogenous, determined by how much effort a project inherently attracts weighted against its value to the manager. I provide a simple condition allowing us to predict how projects are ranked in the optimum.

Definition 6. The manager's valuation is **swap-increasing** if there exists an ordering of projects such that for all pairs of projects $p < q$ and every profile of project qualities $Q^p < Q^q$,

$$V(\dots, Q^p, \dots, Q^q, \dots) \leq V(\dots, Q^q, \dots, Q^p, \dots)$$

where only the p-th and q-th arguments are swapped.

Holding other project qualities fixed, the manager prefers higher quality at project p over project q whenever $p < q$. The log-linear Cobb-Douglas valuation in Section 3 is swap-increasing in the $\gamma^{(\cdot)}$-weight ordering.

Swap-increasing valuations are insufficient to predict project orderings if workers' preferences are heterogeneous. Because some projects naturally attract high effort from workers, their marginal values could be pushed lower in the optimum. To prevent variations in worker preferences from causing unpredictable project reorderings, I assume uniform worker preferences across projects. Complementarity and cost parameters can remain heterogeneous. At the optimum, top-value projects under a swap-increasing valuation achieve maximum quality and team size.

Proposition 3. *If V is swap-increasing and $\beta_i^p \equiv \beta_i$ for all i, then $Q^1 \geq \cdots \geq Q^m$ and $V^1 \geq \cdots \geq V^m$ in the optimum.*

Proof. See Appendix A.10. □

Finally, obtaining a closed-form solution in the assignment and cap cutoffs requires the most restrictive parametric assumptions. In what follows, I assume the manager's

valuation is log-linear Cobb-Douglas and workers are identical. Exploiting a structural symmetry in the workers allows us to pinpoint how the cap cutoff responds to changes in worker attributes.

Proposition 4. *Assume that all workers are identical and the manager's valuation is* $V = \sum_{p=1}^{m} \gamma^{(p)} \ln(Q^p)$ *where* $\gamma^{(1)} \geq \cdots \geq \gamma^{(m)}$ *and* $\sum_{p=1}^{m} \gamma^{(p)} = 1$. *There exists a worker-symmetric optimal project assignment with the following properties.*

1. *Every worker is assigned to every project.*
2. *There exists a single project index* ϕ *where every worker is uncapped at projects* $s \leq \phi$ *and capped at projects* $t > \phi$.
3. *The* ϕ *index is given by*

$$\phi := \max \left\{ \hat{\phi} \;\; s.t. \;\; \frac{\sum_{p=1}^{\hat{\phi}} \gamma^{(p)}}{\gamma^{(\hat{\phi})}} - \hat{\phi} \leq \frac{c - r - g(n-1)}{r} \right\}$$

4. *The project index* ϕ *is non-increasing in* r, g *and* n, *non-decreasing in* c *and independent of* β.

Proof. See Appendix A.11. □

Because the log-linear Cobb-Douglas satisfies Inada conditions at each project, every worker must be assigned to every project in the worker-symmetric optimum. The assignment margin shuts down, allowing us to isolate and focus on whether the manager caps or not. It is characterized by a single project index ϕ which has a closed-form solution.

All else equal, raising multi-tasking costs r or collaborative gains g makes it more valuable for the manager to concentrate more effort on a smaller set of projects. When workers struggle to work on many projects simultaneously, the manager caps them on more projects, pushing them to exert higher effort on more valuable projects. When there are higher gains to collaboration, the manager extracts higher quality from more valuable projects at the expense of the less valuable ones. An increase in team size n likewise strengthens complementarities, and so it operates in the same way as an increase in g. A rise in c makes specialization more expensive, leading the manager to seek a more balanced project assignment. Finally, ϕ does not depend on β because β does not affect the cost of spreading or concentrating effort across projects.

6 Sorting Specialists and Generalists

Which workers specialize in top-ranked projects, and which operate as generalists? Which workers are left uncapped at a project? In general, the ordering of the assignment and cap cutoffs is endogenous, and endogeneity arises from two different sources. The first is which collaborators a worker works with on a particular project. Whether to assign and/or keep a worker uncapped at a project depends on which other workers are staffed there and so each worker's cutoff depends on every other worker's cutoffs. The second is how many projects a worker is on. Because a worker incurs multitasking costs, whether the manager should assign him to one more project depends on how thinly he is already spread. However, the latter is precisely what his assignment and cap cutoffs determine.

To understand how each force— complementarity and effort costs— sorts workers, I shut down each source of endogeneity in turn. In Section 6.1, I assume complementarities are homogeneous, so that the identity of a collaborator is irrelevant to a worker's marginal benefit. I examine how heterogeneity in costs orders workers and derive a statistic in the primitives— the *specialization index*— that orders both workers' assignment and cap cutoffs. In Section 6.2, I allow for heterogeneous complementarity but assume the manager assigns every worker to every project in the optimum. The *scope index*, which measures a worker's comparative advantage as a generalist over a specialist, orders workers' cap cutoffs. I leave the complete discussion of the general endogenous sorting index to Appendix D.

To be clear, I do not make any parametric assumptions on the manager's valuation function in this section. I also assume worker preferences are heterogeneous. As established in Section 5.4, while preferences influence the ordering of projects in the optimum, they do not affect the relative ordering of ϕ_i and α_i across workers. Variations in preferences do not alter the cost of concentrating or spreading a worker across multiple projects.

6.1 The Specialization Index

I assume workers differ in their specialization costs c_i, multitasking costs r_i and preferences β_i^p but have identical complementarity i.e. $g_{ij} \equiv g$ for all $i \neq j$. In this environment, the trade-off inherent to assigning and keeping a worker uncapped at

multiple projects is captured by the *specialization index*.

Definition 7. The **specialization index** of worker i is $\chi_i := r_i/(c_i + g)$.

The index reflects the relative cost of multitasking to specialization. A worker has a high specialization index if multitasking is expensive relative to specialization, and a low index in the reverse case. The following result shows that it sufficiently sorts generalists and specialists.

Theorem 2. *In the optimum, cutoffs for assignment and leaving a worker uncapped α_i and ϕ_i are sorted such that $\chi_i > \chi_j$ implies $\alpha_i \leq \alpha_j$ and $\phi_i \leq \phi_j$.*

Proof. See Appendix A.12. □

In line with intuition, spreading a worker who is a poor multitasker but excels at concentrating on a single project across multiple project teams is not optimal. The manager keeps him specialized on the top-ranked projects in the optimum. The opposite is true for workers who are skilled at multitasking but not at concentrating. The least valuable projects are staffed only by workers with the lowest specialization indices. Specialists, those who are assigned to few projects, work together on large project teams, while generalists may be the only ones employed at a particular project. Holding one cost dimension fixed isolates the effect of the other on assignments and cap cutoffs.

Corollary 1. *The following statements follow directly from Theorem 2.*

1. *If workers have homogeneous specialization costs $c_i \equiv c$, then $r_i > r_j$ implies $\alpha_i \leq \alpha_j$ and $\phi_i \leq \phi_j$.*
2. *If workers have homogeneous multitasking costs $r_i \equiv r$, then $c_i > c_j$ implies $\alpha_i \geq \alpha_j$ and $\phi_i \geq \phi_j$.*

While the extensive margin in assignments and caps is ordered, the intensive margin is not. Effort levels are sensitive to the model parameters. On the one hand, a high multitasking cost leads the manager to reduce that worker's efforts across multiple projects, holding the set of assigned projects fixed. On the other hand, the manager seeks to assign such a worker to a *smaller* set of assigned projects, freeing up his capacity to work more. The net effect can go either way.

Example 1. There are two workers i and j and two projects p and q. The model parameters are $\beta = 1$, $c = 0.6$, $g_{ij} = 0.05$ and multitasking costs $r_i = 0.15$ and $r_j = 0.05$. Worker i is the specialist and worker j is the generalist. The manager has a log-linear Cobb-Douglas valuation. I increase the manager's γ weight on project p from 6 to 7 while keeping her weight on project q at 1.

	$\gamma^{(p)} = 6$			$\gamma^{(p)} = 7$		
	p	q	Total	p	q	Total
Worker i (specialist, $r_i = 0.15$)	1.662	$0.577^{\dagger}$	2.239	1.724	$0.329^{\dagger}$	2.053
Worker j (generalist, $r_j = 0.05$)	1.674	1.575	3.249	1.681	1.554	3.235
Gap $e_j^p - e_i^p$	$+\mathbf{0.012}$			$-\mathbf{0.044}$		

† Capped. Worker i is capped at project q; worker j is uncapped everywhere.

Table 5

The intensive margin is not ordered while the extensive margin is unchanged across the two columns. The generalist is uncapped at both projects, the specialist only at p, yet the ranking of effort levels at p reverses. When $\gamma^{(p)} = 6$, the generalist works more on p; when $\gamma^{(p)} = 7$, the specialist does. The assignment pattern stays the same. The total effort column reveals why this is true. When the manager is more biased towards project p, she tightens the specialist's cap at project q. His total effort falls, dragging his marginal cost down, which frees up his capacity to work more on project p. Since the generalist is uncapped at both projects, the manager has less room to move his efforts around. Whether a specialist works more or less than a generalist on a shared project depends on the levels of the parameters, and not the inherent structure of the assignment.

6.2 The Scope Index

I assume the manager assigns every worker to every project in the optimum and every attribute is heterogeneous, including complementarity. I examine how workers' cap cutoffs are sorted and a natural conjecture is that central workers should be uncapped at many projects. Because they collaborate well with other workers, allowing them to contribute as much as they wish at many projects amplifies efforts throughout the entire organization. However, doing so backfires because multitasking weakens

their contributions, causing other workers to pull back due to a loss in complementarity. A central worker amplifies (1) effort at every project and (2) effort at a single project holding fixed total efforts. To quantify his comparative advantage of (1) over (2) requires comparing two Katz–Bonacich centralities using two previously defined matrices: the aggregate interaction matrix $\mathbf{P}$ and the team interaction matrix $\mathbf{M}$.

Definition 8. The **scope index** of worker i is $\rho_i := \frac{(\mathbf{P}^{-1}\mathbf{1}_n)_i}{(\mathbf{M}^{-1}\mathbf{1}_n)_i}$.

I show in Appendix A.1, both $\mathbf{P}^{-1}$ and $\mathbf{M}^{-1}$ are nonnegative under Assumption 1. Thereby, the scope index is also nonnegative. Recall the matrix definitions

$$\mathbf{P} := \mathbf{C} + (m-1)\mathbf{R} - \mathbf{G} \qquad \mathbf{M} := \mathbf{C} - \mathbf{R} - \mathbf{G}$$

The crucial difference between them lies in their diagonals: $\mathbf{M}$ has $c_i - r_i$ while $\mathbf{P}$ has $c_i + (m-1)r_i$. In particular, the sign of the multitasking cost makes the numerator and denominator behave differently.

The numerator, which is $(\mathbf{P}^{-1}\mathbf{1}_n)_i$, measures the aggregate increase in effort at every project across all workers resulting from a marginal increase in worker i's preferences for working on all projects. Notice that the marginal cost of total effort, $c_i + (m-1)r_i$, grows with multitasking costs. Therefore, workers with low r_i, along with others with strong links to them either directly or through a chain of peers, generate a larger aggregate effort response reflected in a higher $(\mathbf{P}^{-1}\mathbf{1}_n)_i$ value.

Being skilled at multitasking has the opposite effect on the denominator. The entry $(\mathbf{M}^{-1}\mathbf{1}_n)_i$ measures the total increase in effort across all workers on a single project given a marginal increase in worker i's preference there *holding their total efforts fixed*. Having a low multitasking cost means that it is more expensive to redirect efforts towards project p, which has a net marginal cost of $c_i - r_i$. These workers, along with their collaborators, therefore generate a smaller effort response within a project and have a lower $(\mathbf{M}^{-1}\mathbf{1}_n)_i$ value.

Put together, the scope index measures the advantage a worker has as a generalist versus a specialist. Workers with a high scope index, those with relatively high $(\mathbf{P}^{-1}\mathbf{1}_n)_i$ and low $(\mathbf{M}^{-1}\mathbf{1}_n)_i$ values, are better generalists. It is more beneficial to leave them uncapped on a wide range of projects rather than specialize them on a handful of projects. The converse is true for workers with low scope indices, as formalized in the following result.

Theorem 3. *In an optimal project assignment where every worker is assigned to every project, the cap cutoffs are sorted such that $\rho_i > \rho_j$ implies $\phi_i \geq \phi_j$.*

Proof. See Appendix A.13. □

Example 2. Sorting by centralities in the aggregate and within-project interaction matrices alone can get the designation of generalists and specialists. Consider the following example with three workers and three projects. Workers have identical costs and preferences $c_i = 1$, $r_i = 0.2$ and $\beta_i^p = 1$ and the manager's valuation is given by $V = 3\ln(Q^1) + 2\ln(Q^2) + \ln(Q^3)$. The complementarities are given by $(g_{12}, g_{13}, g_{23}) = (0.20, 0.15, 0.02)$.

	$(\mathbf{P}^{-1}\mathbf{1}_n)_i$	$(\mathbf{M}^{-1}\mathbf{1}_n)_i$	ρ	Project		
				1	2	3
Worker 1	**0.93**	**2.01**	**0.46**	Uncapped	Uncapped	**Capped**
Worker 2	0.86	1.79	0.48	Uncapped	Uncapped	Uncapped
Worker 3	0.83	1.67	0.49	Uncapped	Uncapped	Uncapped

Every worker is assigned to every project. However, worker 1, who is the most central worker on *both* centrality measures is the one who is capped on project 3.

Once the assumption that every worker is assigned to every project is dropped, the sorting index becomes endogenous. Two sets of multipliers must be solved jointly: (1) those on the incentive constraints, which govern caps and (2) those on the nonnegativity constraints, which govern assignment. Under heterogeneous complementarity (even under homogeneous costs), neither admits a closed form. Appendix D develops the resulting index.

7 Comparative Statics

That the incentive constraints are linear in efforts permits us to interpret them as linear budgets over workers' efforts. The comparative statics admit a Slutsky decomposition into a substitution and income effect. The substitution effect accounts for how the manager substitutes one worker's effort at a project for another if the former

becomes more costly to use. The income effect further decomposes into two separate network effects.

To isolate the income effect, I examine the comparative statics of optimal project assignments that leaves every worker uncapped everywhere. That is, a worker's incentive constraint binds at a project whenever he is assigned there, and I show that these assignments have zero substitution effect. Mathematically, differentiating the linear system in the incentive constraints alone delivers the comparative statics. The income effect separates into a *within-project spillover* and a *network reallocation effect*. The former operates through effort complementarity and is determined by the team interaction matrix $\mathbf{M}$, while the latter reflects how multitasking induces workers to redistribute effort across projects, as dictated by the aggregate interaction matrix $\mathbf{P}$.

7.1 A Slutsky Decomposition

The comparative statics assume that given a shock, assignment and cap patterns do not change in the optimum. That is, I focus on a *local* analysis holding fixed $A := \{(i,p) \mid e_i^p > 0\}$ and $\Phi := \{(i,p) \mid i \text{ is uncapped at } p\}$. I interpret a binding incentive constraint as a linear effort budget.

$$\beta_i^p = c_i e_i^p + r_i \sum_{q \neq p} e_i^q - \sum_{j \neq i} g_{ij} e_j^p$$

The budget reading is as follows: β_i^p is the income of effort available to the manager, c_i is the own price of worker i's effort at project p, r_i is the price of his effort at other projects and $-g_{ij}$ is a discount because worker j's effort makes it cheaper for worker i to work more.

I introduce the *substitution* and *income matrices*. Recall that the worker interaction matrix $\mathbf{W}$ encodes each incentive constraint as a row; the relevant submatrix is $\mathbf{W}_\Phi$ with rows restricted to worker-project pairs that are left uncapped. At the optimum, the efforts and multipliers $(\mathbf{e}_A, \boldsymbol{\lambda}_\Phi)$ satisfy the Kuhn-Tucker conditions:

$$\begin{aligned} \nabla_{\mathbf{e}} V(\mathbf{e}_A) - \mathbf{W}_\Phi^\intercal \boldsymbol{\lambda}_\Phi &= \mathbf{0} \\ \boldsymbol{\beta}_\Phi - \mathbf{W}_\Phi \mathbf{e}_A &= \mathbf{0} \end{aligned}$$

Implicit differentiation of this system yields the following block Jacobian matrix.

Definition 9. The Jacobian matrix $\mathbf{J}$ and its inverse (if it exists) written in corresponding block matrix form are

$$\mathbf{J} := \begin{bmatrix} \nabla^2_{\mathbf{e}} V(\mathbf{e}_A) & -\mathbf{W}_\Phi^\intercal \\ -\mathbf{W}_\Phi & \mathbf{0} \end{bmatrix} \qquad \mathbf{J}^{-1} := \begin{bmatrix} (\mathbf{J}^{-1})_{11} & (\mathbf{J}^{-1})_{12} \\ (\mathbf{J}^{-1})_{21} & (\mathbf{J}^{-1})_{22} \end{bmatrix}$$

The **substitution matrix** is $\mathbf{S} := (\mathbf{J}^{-1})_{11}$ and the **income matrix** is $\mathbf{Y} := -(\mathbf{J}^{-1})_{12}$.

The substitution matrix $\mathbf{S}$ collects every direction along which the manager can reallocate effort while holding her "purchasing power" of effort fixed. It satisfies $\mathbf{W}_\Phi \mathbf{S} = \mathbf{0}$, and so any reallocation along $\mathbf{S}$ leaves every binding incentive constraint binding as before. This is the effort analog to a movement along a budget line. Moreover, $\mathbf{S}$ is symmetric and negative semidefinite, the latter of which reflects the law of compensated demand.

The income matrix $\mathbf{Y}$ accounts for directions that move the constraints themselves. It satisfies $\mathbf{W}_\Phi \mathbf{Y} = \mathbf{I}$. Each of its columns corresponds to a direction that shifts a binding constraint by exactly one unit and keeps every other binding constraint undisturbed. This is the effort analog of a parallel shift of the budget line after income changes, holding relative prices fixed.

The Slutsky decomposition requires a technical condition to ensure the inverse of the Jacobian $\mathbf{J}$ exists.

Theorem 4. *If $\nabla^2_e V(\mathbf{e}_A)$ is strictly negative definite on $\ker(\mathbf{W}_\Phi)$, then $\mathbf{J}^{-1}$ exists and at the optimum $(\mathbf{e}_A, \boldsymbol{\lambda}_\Phi)$ and for every parameter θ,*

$$\frac{\partial \mathbf{e}_A}{\partial \theta} = \underbrace{\mathbf{S}\left(\frac{\partial \mathbf{W}_\Phi^\intercal}{\partial \theta}\boldsymbol{\lambda}_\Phi\right)}_{\textit{substitution}} + \underbrace{\mathbf{Y}\left(\frac{\partial \boldsymbol{\beta}_\Phi}{\partial \theta} - \frac{\partial \mathbf{W}_\Phi}{\partial \theta}\mathbf{e}_A\right)}_{\textit{income}}$$

Proof. See Appendix A.14. □

The condition is sufficient for the optimum to be unique in individual efforts so that the matrix $\mathbf{J}$ is full rank. While $\mathbf{S}$ and $\mathbf{Y}$ represent the directions of the substitution and income effects to which the manager responds, the magnitude of each response is captured by how a shock affects the incentive constraints through $\frac{\partial \mathbf{W}_\Phi}{\partial \theta}$ (and its transpose). To simplify notation, I derive the formulas for $\frac{\partial \mathbf{W}}{\partial \theta}$ which I call the **impact vector**; the corresponding expressions for $\mathbf{W}_\Phi$ are the rows of the following matrices restricted

to indices in Φ.

$$\frac{\partial \mathbf{W}}{\partial \beta_i^p} = \mathbf{0} \qquad \frac{\partial \mathbf{W}}{\partial c_i} = (\boldsymbol{\epsilon}_i \boldsymbol{\epsilon}_i^\intercal) \otimes \mathbf{I}_m$$

$$\frac{\partial \mathbf{W}}{\partial g_{ij}} = -(\boldsymbol{\epsilon}_i \boldsymbol{\epsilon}_j^\intercal + \boldsymbol{\epsilon}_j \boldsymbol{\epsilon}_i^\intercal) \otimes \mathbf{I}_m \qquad \frac{\partial \mathbf{W}}{\partial r_i} = (\boldsymbol{\epsilon}_i \boldsymbol{\epsilon}_i^\intercal) \otimes (\mathbf{1}_m \mathbf{1}_m^\intercal - \mathbf{I}_m)$$

where $\boldsymbol{\epsilon}_i$ is the i-th basis vector and $\otimes$ is the Kronecker product.

To see how this Slutsky decomposition compares to that of traditional consumer theory, consider an increase in specialization cost for worker i. Recall ϕ_i denotes the last project worker i is uncapped at.

$$\frac{\partial \mathbf{e}_A}{\partial c_i} = \sum_{p=1}^{\phi_i} \left(\mathbf{S}_{\cdot,(i,p)} \lambda_i^p - \mathbf{Y}_{\cdot,(i,p)} e_i^p \right)$$

The manager is the consumer, but she faces a separate budget per binding incentive constraint rather than a single one. A price change is usually tracked in one budget, and the income effect is weighted by the quantity of the good whose price moved. Here, c_i is the price of worker i's effort at every project he is uncapped, and a change in c_i moves all ϕ_i budget lines at the same time.

7.2 Decomposing the Income Effect

There is a class of project assignments with no substitution effect. When workers are uncapped whenever they are assigned, any reallocation of efforts along one incentive constraint violates other incentive constraints. Geometrically, the feasible set, which is defined by linear constraints, forms a polytope. Caps place the optimum on a face of the polytope and the substitution effect corresponds to movements across that surface.

If the optimum lands on a vertex of the polytope ($\hat{e}$ in Figure 5), movement along an incentive constraint necessarily moves off another binding constraint. These are project assignments where every worker is uncapped at every assigned project.

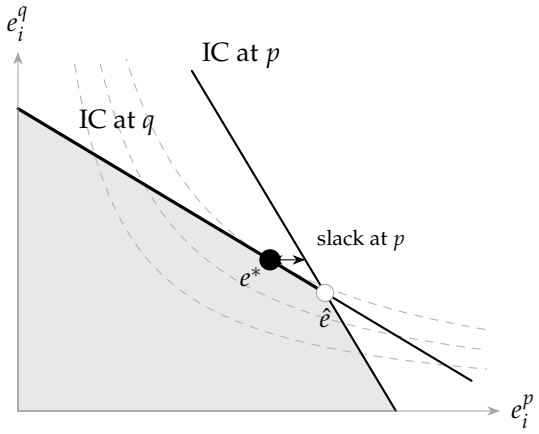


Figure 5: Capped at $\mathbf{e}^*$ for one worker and two projects.

Proposition 5. *If every worker assigned to a project is uncapped in the optimum, then* $\mathbf{S} = \mathbf{0}$.

Proof. See Appendix A.15. □

By isolating the income effect in projects where every worker is uncapped, I demonstrate that it can be further decomposed into two separate components: (1) a *within-project spillover* where workers redirect effort towards a project receiving a positive shock and (2) a *network reallocation effect*, where workers adjust effort across different projects to manage multitasking costs.

If the manager assigns and leaves every worker uncapped at every project, then the assignment is given by the no-intervention outcome $\hat{\mathbf{e}} = \mathbf{W}^{-1}\boldsymbol{\beta}$. The income matrix is $\mathbf{W}^{-1}$.[14] To compute the comparative statics at the project-level, I define project-specific impact vectors.

Definition 10. For parameter $\theta \in \{\beta_j^q,\ c_j,\ r_j,\ g_{jk}\}$, $\mathbf{d}^p(\theta) \in \mathbb{R}^n$ is the **impact vector at project** p and is given by

$$\mathbf{d}^p(\beta_j^q) = \delta_{pq}\boldsymbol{\epsilon}_j \qquad \mathbf{d}^p(c_j) = -e_j^{p*}\boldsymbol{\epsilon}_j$$
$$\mathbf{d}^p(g_{jk}) = e_k^{p*}\boldsymbol{\epsilon}_j + e_j^{p*}\boldsymbol{\epsilon}_k \qquad \mathbf{d}^p(r_j) = -(E_j^* - e_j^{p*})\boldsymbol{\epsilon}_j$$

where δ_{pq} is the Kronecker delta, $\boldsymbol{\epsilon}_j$ is the j-th standard basis vector and E_j is worker j's total effort.

[14] Reading off Theorem 4, we have $\frac{\partial \mathbf{e}}{\partial \theta} = \mathbf{W}^{-1}\left(\frac{\partial \boldsymbol{\beta}}{\partial \theta} - \frac{\partial \mathbf{W}}{\partial \theta}\mathbf{e}\right)$.

Splitting the income effect into the two network effects requires a matrix decomposition of the worker interaction matrix into the familiar matrices $\mathbf{M}^{-1}$ and $\mathbf{P}^{-1}$.

Proposition 6. *The following decomposition holds*

$$\mathbf{W}^{-1} = \mathbf{M}^{-1} \otimes \mathbf{I}_m - (\mathbf{M}^{-1}\mathbf{R}\mathbf{P}^{-1}) \otimes \mathbf{1}_m \mathbf{1}_m^\intercal.$$

This implies that in an optimal project assignment that assigns and leaves every worker uncapped at every project, the following holds for every parameter θ and every project p

$$\frac{\partial \mathbf{e}^p}{\partial \theta} = \underbrace{\mathbf{M}^{-1}\left(\mathbf{d}^p(\theta)\right)}_{\textit{within-project spillover}} - \underbrace{\mathbf{M}^{-1}\mathbf{R}\mathbf{P}^{-1}\textstyle\sum_{q=1}^m \mathbf{d}^q(\theta)}_{\textit{network reallocation}}$$

Moreover $\frac{\partial e_i^p}{\partial \beta_j^p} > 0$ *for all* j *and* $\frac{\partial e_i^p}{\partial \beta_j^q} < 0$ *for all* $q \neq p$.

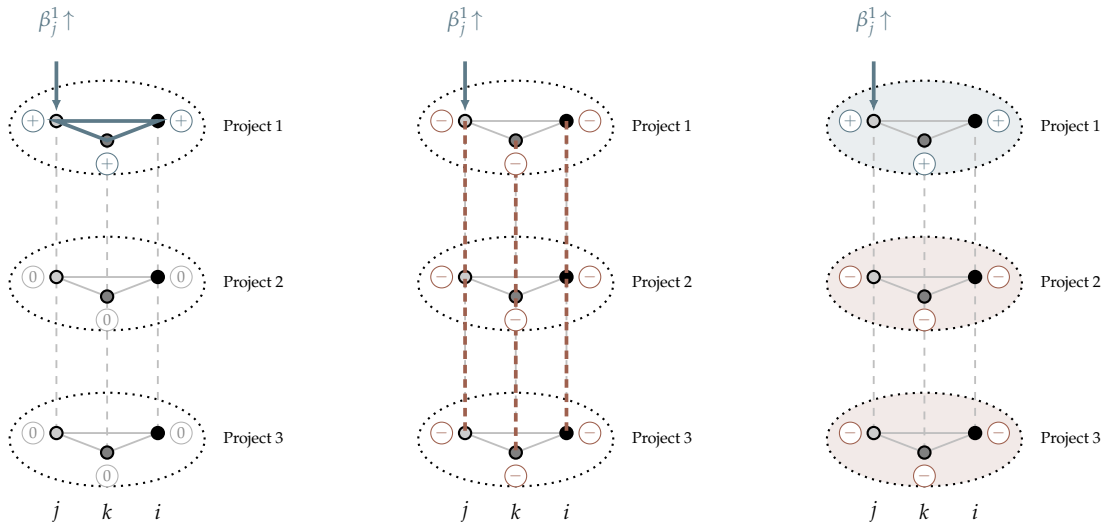
Proof. See Appendix A.16. □

To see the two network effects in action, I focus on how the manager adjusts worker i's effort at project p given a positive preference shock for worker j at project q.

$$\frac{\partial e_i^p}{\partial \beta_j^q} = \underbrace{\mathbf{M}_{ij}^{-1}\delta_{pq}}_{\text{within-project spillover}} - \underbrace{\sum_{\ell=1}^n \mathbf{M}_{i\ell}^{-1} r_\ell \mathbf{P}_{\ell j}^{-1}}_{\text{network reallocation}}$$

The first term is local to the project experiencing the shock. Worker j is willing to work more there, and complementarity makes everyone else supply more effort. The $\mathbf{M}_{ij}^{-1}$ entry counts the total complementarity effect for worker i at project q while total efforts are held fixed.

The second term represents a separate reallocation channel on the entire network. The $\mathbf{P}_{\ell j}^{-1}$ entry captures how much worker ℓ's total effort rises in response to an increase in worker j's preference for working on any project. Multiplying by r_ℓ converts that extra effort into an increase in worker ℓ's marginal cost, which applies at every project he works on. In response, he reduces effort at every project, and $\mathbf{M}_{i\ell}^{-1}$ propagates this back through the within-project network to worker i: he experiences a loss in the complementarity from worker ℓ. The network reallocation effect causes a drop in quality for every project not subject to the positive preference shock. The following figure illustrates.



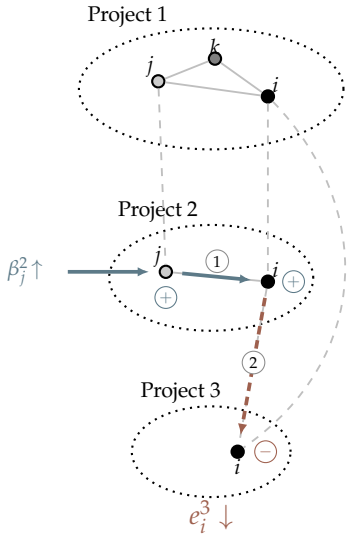
(a) Within-project spillover. (b) Network reallocation. (c) Net response.

Figure 6: Nodes of the same color correspond to the same worker. The ellipses represent different projects. Worker j receives a preference shock at project 1.

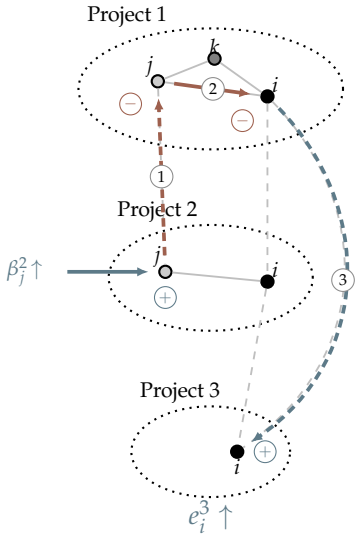
What changes when the manager does not assign every worker to every project? The magnitude and direction of a shock transmission becomes influenced by the pattern of active project assignments. The following example illustrates.

Example 3. Suppose there are three projects $(1, 2, 3)$ and three workers (i, j, k), and the optimal project assignment has the following nested structure as depicted in Figure 7: worker i is assigned to all 3 projects, worker j is assigned to projects 1 and 2 and worker k is assigned to project 1.

Consider an increase in worker j's preference for working on project 2, denoted by $\beta_j^{(2)}$. This shock propagates through two distinct channels. Worker i raises his efforts at project 2 due to an increase in complementarity, and cuts back on his efforts at project 3. The other channel sees worker j withdraw from project 1. Worker i follows suit at project 1, saving capacity to work on his other projects. This effect pushes worker i to *raise* his effort at project 3. The net direction of the shock is ambiguous and hinges on the strength of his effort complementarity. For example, if g_{ij} is large, then his gain in effort at project 2 dominates, leading to higher total effort and an overall contraction at project 3. On the other hand, if g_{ij} is low, worker i's effort reduction at project 1 could dominate and he works more on project 3. In contrast to the full assignment case, a positive preference shock for one project could *raise* the quality of another project.



(a) The direct route $(j,2) \to (i,2) \to (i,3)$. Complementarity raises worker i's effort at project 2; his multitasking cost then pulls him off project 3.



(b) The indirect route, $(j,2) \to (j,1) \to (i,1) \to (i,3)$. Worker j reduces effort at project 1, worker i loses complementarity, and he reduces effort at project 1. This frees up effort for i to spend at project 3.

Figure 7: Two routes for the same shock under partial assignment.

Under partial assignment, the project impact vectors remain unchanged. The primary distinction lies in the team interaction matrix. Within-project spillovers for project p are governed by $\mathbf{M}^{(A^p)}$, the principal submatrix of $\mathbf{M}$ restricted to workers assigned to p. Likewise, the network reallocation effect changes as well because each project carries a different $\mathbf{M}^{(A^p)}$ matrix. The operator that captures how total efforts respond to the shock is no longer $\mathbf{P}$, but $\mathcal{P}$, which depends on the pattern of assignment. It has entries

$$\mathcal{P}_{ij} = \delta_{ij} + r_j \sum_{p \in P_i \cap P_j} (\mathbf{M}^{(A^p)})^{-1}_{ij}.$$

where P_i is the set of projects worker i is assigned to.

Proposition 7. *In an optimal project assignment with partial assignment and leaves every worker uncapped at every assigned project, the following holds for every parameter θ and every project p*

$$\frac{\partial e^p}{\partial \theta} = \underbrace{(\mathbf{M}^{(A^p)})^{-1}\mathbf{d}^p(\theta)}_{\textit{within-project spillover}} - \underbrace{(\mathbf{M}^{(A^p)})^{-1}\mathbf{R}\big(\mathcal{P}^{-1}\boldsymbol{\nu}(\theta)\big)}_{\textit{network reallocation}}$$

where the right-hand vector $\boldsymbol{\nu}$ has entries

$$\boldsymbol{\nu}_i(\theta) := \sum_{p \mid i \in A^p} \left((\mathbf{M}^{(A^p)})^{-1} \mathbf{d}^p(\theta) \right)_i$$

Proof. See Appendix A.17. □

The comparative statics are carried by the assignment-specific interaction matrices.[15] As the example demonstrated, a positive preference shock could raise the qualities of other projects; imbalanced project assignments lead to different routes the shock can take.

8 Discussion

8.1 Effort Caps, Contracts and Floors

By studying optimal project assignments via effort caps in isolation, I prohibit the manager from contracting on effort or setting effort floors. A natural question is whether she could achieve the outcomes if allowed to do so. The comparison here is about implementable outcomes, not which instrument the manager would prefer as the main analysis concerns environments where contracting and effort floors are unavailable. Answering the latter requires additional assumptions on how her valuation of projects relates to the costs of paying her workers. I begin with discussion of contracts and show that effort caps implement effort profiles that contracts which reward effort cannot.

Proposition 8. *Suppose worker i's payoff under contract w_i is given by $U_i(\mathbf{e}_i, \mathbf{e}_{-i}) + w_i(\mathbf{e}_i, \mathbf{e}_{-i})$ where w_i is nondecreasing in each e_i^p. No effort profile where worker i is capped at project p is an equilibrium.*

Proof. See Appendix A.18. □

Any implementing contract must reduce payment enough to offset the worker's utility gain from sufficiently small increases in effort above the target. If the contract is

[15] One can readily verify that Proposition 6 is a special case of the above: under full assignment $(\mathbf{M}^{(A^p)})^{-1} \equiv \mathbf{M}^{-1}$, so $\mathcal{P} = \mathbf{I} + m\mathbf{M}^{-1}\mathbf{R} = \mathbf{M}^{-1}\mathbf{P}$ and $\boldsymbol{\nu}(\theta) = \mathbf{M}^{-1} \sum_{q=1}^{m} \mathbf{d}^q(\theta)$.

differentiable, its marginal reward at the target must be negative. For an illustration, consider a linear contract for each worker i

$$w_i(\mathbf{e}_i) = \sum_{p=1}^{m} b_i^p e_i^p \text{ where } b_i^p \in \mathbb{R}.^{16}$$

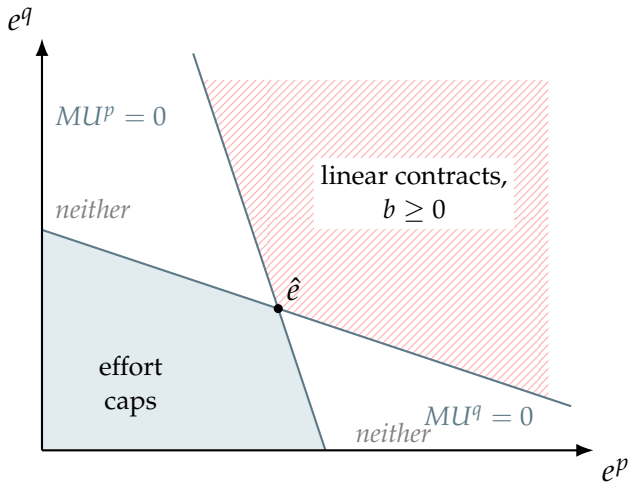


Figure 8: Effort profiles implementable by caps and by contracts, for one worker and two projects. The lines are the worker's incentive constraints. The two sets meet only at the unconstrained equilibrium $\hat{e}$. In the two unshaded regions, neither instrument alone suffices.

To implement an effort profile, the required rate is the negative of the worker's marginal utility at the target. To cap a worker and leave his incentive constraint slack, the manager must fine him for working. Effort caps and *nonnegative* wages are different instruments that move effort in the opposite direction. A cap holds a worker back on a project he would like to contribute more to, while a positive wage pushes him to do so. Instead, contracts can achieve effort profiles where worker marginal utilities are negative, something effort caps cannot accomplish.

Similar to contracts that reward effort, effort floors cannot implement profiles in which a worker is capped. A floor cannot stop a worker from exceeding a specific effort level, so a worker will always exert more effort if he desires. Floors instead push workers above the effort they would otherwise contribute. Without an outside option for workers, nothing limits how far the manager can push them. In fact, she would

[16] I leave out a fixed payment because it does not affect workers' incentives to exert effort. It affects whether limited liability binds and can be chosen so that total pay is always nonnegative.

push their effort floors without bound, so the manager's problem is well posed only if workers can quit or face institutional limits on their total effort.

8.2 Asymmetric Complementarities

I imposed symmetric complementarities early on to write the workers' effort game as an exact potential game, which buys us uniqueness of equilibrium. Beyond that, complementarities can be asymmetric, and the main structural results still hold.

It is imperative that worker interactions are complements and not substitutes within a project. If interactions are substitutes i.e. $g_{ij} < 0$, then the team-coordinated effort transfer argument is no longer available. Shifting a worker's effort from a low-value project to a high-value one makes his teammates desire less effort at the latter, potentially rendering the transfer infeasible. The nested hierarchy structure is no longer guaranteed.

8.3 Effort Budgets

Suppose there is a maximum level of worker effort that the manager can implement in equilibrium e.g. a 40-hour work week. The restriction places an additional constraint in the manager's convex optimization problem where, for all workers i, $\sum_{p=1}^{m} e_i^p \leq T_i$, and T_i is an exogenously given hard limit. Workers' preferences and choice problems remain unchanged. The team-coordinated transfer keeps workers' total efforts constant, so the argument establishing nested team membership remains available. Theorem 1 does not carry over verbatim because the effort budget could be exhausted before incentive constraints for a worker bind. His initial interval of projects where he is left uncapped may therefore be empty and so this extension needs to allow $\phi_i = 0$ could be 0. The sorting results are not robust. For example, even if a worker has a high scope index, implying they should be left uncapped across many project teams, they may operate with a very low T_i. Consequently, the manager cannot leave them uncapped in as many teams.

9 Concluding Remarks

Organizations do not pay for effort on a day-to-day basis. Instead, they choose how much they want implemented by assignments. Effort caps are not a substitute for incentives and deserve their own analysis. They are essentially two instruments in one: they select which workers collaborate with whom, and dictate the scope of what a worker works on in the organization. Optimally assigning workers to projects by setting effort caps turns out to have a much more structured solution than the richness of the model suggests. A single endogenous ranking of projects sorts every worker's assignment into three consecutive intervals— assigned and uncapped, assigned but capped, unassigned. I showed that centrality alone does not sort specialists from generalists. Instead, a ratio of two Katz–Bonacich centralities sorts workers by weighing the value of strong ties within a team and against the cost of spreading it across many project teams. Because the assignment selects the network of interactions, an organization's response to shocks is implicitly designed by the manager. A shock favorable to one project could end up being bad news for other projects in the organization, depending on the pattern of assignment and caps in the optimum.

There are many promising extensions for future research. Because this paper studies assignment in isolation, a natural next step is to study the joint design of assignments and incentives, especially given that they affect effort in opposite ways. Project assignments could also be studied in tandem with career concerns, as success on certain projects could advance a worker's career more than others. Another direction is to analyze the welfare effects of effort caps from a social planner's perspective. Lastly, Assumption 1 guarantees a unique equilibrium and nullifies equilibrium selection concerns. Relaxing it allows us to study which project assignments remain robust when the worst equilibrium is played. I hope that this work inspires future research in studying staffing decisions and project assignment through the lens of a network design problem.

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A Omitted Proofs

A.1 Proof of Lemma 1

To characterize effort profiles that are implementable as equilibria, I exploit symmetric effort complementarity to cast the workers' effort game as an exact potential game. Nash equilibria of the effort game correspond to maxima of the exact potential (Monderer and Shapley, 1996), which is given by

$$\Psi(\mathbf{e}) := \sum_{i=1}^{n} U_i(\mathbf{e}) - \frac{1}{2}\sum_{i=1}^{n}\sum_{j\neq i}\sum_{p=1}^{m} g_{ij} e_i^p e_j^p.$$

If the exact potential is strictly concave and has a unique maximizer corresponding to a unique Nash equilibrium. I show that Assumption 1 implies the Hessian of the exact potential is negative definite, which is given by

$$\mathbf{H} := -(\mathbf{C}\otimes\mathbf{I}_m + \mathbf{R}\otimes(\mathbf{1}_m\mathbf{1}_m^\intercal - \mathbf{I}_m)) + (\mathbf{G}\otimes\mathbf{I}_m)$$

where $\otimes$ is the Kronecker product.[17] For an effort profile $\mathbf{e}$, let $\mathbf{e}^p$ denote the vector of efforts at project p and $E_i := \sum_p e_i^p$. Since $\sum_{p\neq q} e_i^p e_i^q = E_i^2 - \sum_p (e_i^p)^2$, we have

$$\mathbf{e}^\intercal\mathbf{H}\mathbf{e} = -\sum_{p=1}^{m}(\mathbf{e}^p)^\intercal(\mathbf{C}-\mathbf{R}-\mathbf{G})\mathbf{e}^p - \sum_{i=1}^{n} r_i E_i^2$$

It suffices to show that $\mathbf{M} := \mathbf{C}-\mathbf{R}-\mathbf{G}$ is positive definite. Note that $\mathbf{C}-\mathbf{R}$ is a positive diagonal matrix since we assumed that $c_i > r_i$. Set $\mathbf{S} := (\mathbf{C}-\mathbf{R})^{-1/2}\mathbf{G}(\mathbf{C}-\mathbf{R})^{-1/2}$. Observe that $(\mathbf{C}-\mathbf{R})^{-1}\mathbf{G} = (\mathbf{C}-\mathbf{R})^{-1/2}\mathbf{S}(\mathbf{C}-\mathbf{R})^{1/2}$ so $\mathbf{S}$ and $(\mathbf{C}-\mathbf{R})^{-1}\mathbf{G}$ are similar and share the same eigenvalues $\lambda_1, ..., \lambda_n$. These are real because $\mathbf{S}$ is symmetric and satisfy $|\lambda_k| < 1$ by Assumption 1. Therefore, the symmetric matrix $\mathbf{I}-\mathbf{S}$ has eigenvalues $1-\mu_k > 0$, so it is positive definite. Finally, we have $(\mathbf{C}-\mathbf{R})^{1/2}(\mathbf{I}-\mathbf{S})(\mathbf{C}-\mathbf{R})^{1/2} = \mathbf{C}-\mathbf{R}-\mathbf{G} = \mathbf{M}$. For any $\mathbf{x}\neq\mathbf{0}$, the vector $\mathbf{y} := (\mathbf{C}-\mathbf{R})^{1/2}\mathbf{x}$ is nonzero and $\mathbf{x}^\intercal\mathbf{M}\mathbf{x} = \mathbf{y}^\intercal(\mathbf{I}-\mathbf{S})\mathbf{y} > 0$, which completes the argument.

I note a couple of additional useful properties. Because $\mathbf{M}$ is positive definite, sym-

[17] Formally, given a $m\times n$ matrix $\mathbf{A}$ and a $p\times q$ matrix $\mathbf{B}$, the Kronecker product $\mathbf{A}\otimes\mathbf{B}$ is a $mp\times nq$ block matrix given by $\mathbf{A}\otimes\mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ \vdots & \ddots & \vdots \\ a_{m1}\mathbf{B} & \cdots & a_{mn}\mathbf{B}\end{bmatrix}$.

metric with negative off-diagonals, it is an M-matrix and has a nonnegative inverse matrix. Therefore, the matrix $\mathbf{P} := \mathbf{C} + (m-1)\mathbf{R} - \mathbf{G} = \mathbf{M} + m\mathbf{R}$ is likewise an M-matrix with a nonnegative inverse matrix. □

A.2 Proof of Lemma 2

Worker i maximizes $U_i(\mathbf{e}_i, \mathbf{e}_{-i})$, which is concave in own efforts, over $\prod_{p=1}^{m}[0, \bar{e}_i^p]$. Therefore the first-order conditions are necessary and sufficient. His effort $\mathbf{e}_i$ is a best response if and only if for every project p the following holds:

(1) If $0 < e_i^p < \bar{e}_i^p$, then $\frac{\partial U_i}{\partial e_i^p} = 0$.

(2) If $e_i^p = \bar{e}_i^p > 0$, then $\frac{\partial U_i}{\partial e_i^p} \geq 0$.

(3) If $e_i^p = 0 < \bar{e}_i^p$, then $\frac{\partial U_i}{\partial e_i^p} \leq 0$.

(4) If $e_i^p = \bar{e}_i^p = 0$, then no restriction on $\frac{\partial U_i}{\partial e_i^p}$ is required.

(Forward Direction) Suppose $\mathbf{e}$ is a Nash equilibrium given $\bar{\mathbf{e}}$ and assume that $e_i^p > 0$. Since $\bar{e}_i^p \geq e_i^p > 0$, then either Condition (1) or (2) holds. Compute

$$\frac{\partial U_i}{\partial e_i^p} = \beta_i^p + \sum_{j \neq i} g_{ij} e_j^p - c_i e_i^p - r_i \sum_{q \neq p} e_i^q \geq 0 \implies \beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q$$

which is the desired inequality.

(Reverse Direction). Let $\mathbf{e}$ satisfy $\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q$ whenever $e_i^p > 0$ and set $\bar{\mathbf{e}} = \mathbf{e}$. Fix a worker i and a project p. If $e_i^p > 0$, then $e_i^p = \bar{e}_i^p > 0$ and Condition (2) is satisfied because $\frac{\partial U_i}{\partial e_i^p} \geq 0$ by assumption. If $e_i^p = 0$, then $\bar{e}_i^p = 0$ and Condition (4) is vacuously satisfied. The worker has no deviation available is forced to exert zero effort. Every project satisfies the best response conditions, so $\mathbf{e}_i$ is a best response to $\mathbf{e}_{-i}$ and $\mathbf{e}$ is a Nash equilibrium. □

A.3 Proof of Lemma 3

I first show the following lemma.

Lemma 7. *Under Assumption 1, the following statements are true.*

1. *The inverse of* $\mathbf{P}$ *exists and is nonnegative.*
2. *Every effort profile satisfying every incentive constraint satisfies* $\mathbf{E} \leq \tilde{\mathbf{E}} := \mathbf{P}^{-1} \sum_{p=1}^{m} \boldsymbol{\beta}^p$.

Proof. In Appendix A.1, I show that $\mathbf{P}$ is an M-matrix under Assumption 1. Having a nonnegative inverse matrix is a property of M-matrices.

Suppose an effort profile satisfies every incentive constraint. Fix a project p and stack each worker's incentive constraint:

$$\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q \implies \boldsymbol{\beta}^p \geq (\mathbf{C} - \mathbf{R} - \mathbf{G})\mathbf{e}^p + \mathbf{RE}$$

where $\mathbf{E} := [\sum_{p=1}^{m} e_1^p \cdots \sum_{p=1}^{m} e_n^p]$ is the vector of worker total efforts. The implication holds by rewriting $\sum_{q \neq p} e_i^q = \sum_{q=1}^{m} e_i^q - e_i^p$. Finally, aggregate the incentive constraints across all m projects to obtain an inequality in total efforts

$$\sum_{p=1}^{m} \boldsymbol{\beta}^p \geq (\mathbf{C} - \mathbf{R} - \mathbf{G})\mathbf{E} + m\mathbf{RE} \implies \sum_{p=1}^{m} \boldsymbol{\beta}^p \geq \mathbf{PE}.$$

Therefore $\tilde{\mathbf{E}} := \mathbf{P}^{-1} \sum_{p=1}^{m} \boldsymbol{\beta}^p$ is the upper bound on the total efforts. □

Continuing with the main proof, assume that $r_i \tilde{E}_i < \min_p \beta_i^p$. By construction, $\tilde{E}_i$ satisfies

$$(c_i + r_i(m-1))\tilde{E}_i = \sum_{p=1}^{m} \beta_i^p + \sum_{j \neq i} g_{ij} \tilde{E}_j$$

Let $\mathcal{A}_i^{(k)}$ be the set of the k largest entries of $\{\beta_i^p\}_{p=1}^m$. Subtract to obtain

$$(c_i + r_i(m-1))\tilde{E}_i - \sum_{p \in \mathcal{A}_i^{(k)}} \beta_i^p - \sum_{j \neq i} g_{ij} \tilde{\mathbf{E}}_j = \sum_{p \notin \mathcal{A}_i^{(k)}} \beta_i^p \geq (m-k) \min_p \beta_i^p \geq (m-k) r_i \tilde{E}_i$$

Rearrange to get

$$(m-k) r_i \tilde{E}_i \leq (c_i + r_i(m-1))\tilde{E}_i - \sum_{p \in \mathcal{A}_i^{(k)}} \beta_i^p - \sum_{j \neq i} g_{ij} \tilde{E}_j$$

$$\implies (c_i + r_i(k-1))\tilde{E}_i \geq \sum_{p \in \mathcal{A}_i^{(k)}} \beta_i^p + \sum_{j \neq i} g_{ij} \tilde{E}_j \quad (*)$$

Now I show that for any implementable $\mathbf{e}$, the total efforts $E_i := \sum_{p=1}^{m} e_i^p$ are bounded from above by $\tilde{E}_i$. Summing worker i's incentive constraints over projects $p \in P_i$ gives

$$\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq c_i e_i^p + r_i \sum_{q \neq p} e_i^q \implies \sum_{p \in P_i} \beta_i^p + \sum_{j \neq i} g_{ij} \sum_{p \in P_i} e_j^p \geq \sum_{p \in P_i} (c_i e_i^p + r_i \sum_{q \neq p} e_i^q)$$

The cost side simplifies to

$$\sum_{p \in P_i} (c_i e_i^p + r_i \sum_{q \neq p} e_i^q) = (c_i + r_i(|P_i| - 1)) E_i$$

Now suppose that $\zeta := \max_i (E_i / \tilde{E}_i) > 1$. Then multiply and divide by $\tilde{E}_i$ on both the left and right sides to get

$$\zeta(c_i + r_i(|P_i| - 1))\tilde{E}_i \leq \sum_{p \in P_i} \beta_i^p + \zeta \sum_{j \neq i} g_{ij} \tilde{E}_j = \sum_{p \in P_i} \beta_i^p + \sum_{j \neq i} g_{ij} \tilde{E}_j + (\zeta - 1) \sum_{j \neq i} g_{ij} \tilde{E}_j$$

Applying the $(*)$ relation, we have

$$\sum_{p \in P_i} \beta_i^p + \sum_{j \neq i} g_{ij} \tilde{E}_j + (\zeta - 1) \sum_{j \neq i} g_{ij} \tilde{E}_j \leq (c_i + r_i(|P_i| - 1))\tilde{E}_i + (\zeta - 1) \sum_{j \neq i} g_{ij} \tilde{E}_j$$

Chaining the inequality from the two lines above:

$$(\zeta - 1)(c_i + r_i(|P_i| - 1))\tilde{E}_i \leq (\zeta - 1) \sum_{j \neq i} g_{ij} \tilde{E}_j \implies (c_i + r_i(|P_i| - 1))\tilde{E}_i \leq \sum_{j \neq i} g_{ij} \tilde{E}_j$$

This implies that $\sum_{p \in \mathcal{A}_i^{(k)}} \beta_i^p \leq 0$, which is a direct contradiction that $\beta_i^p > 0$ for all i and p.

Now assume $\mathbf{e}$ is implementable with $e_i^p = 0$. Then $\sum_{q \neq p} e_i^q = E_i \leq \tilde{\mathbf{E}}_i$ and so

$$\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p \geq \min_p \beta_i^p > r_i \tilde{\mathbf{E}}_i \geq r_i \sum_{q \neq p} e_i^q$$

Because worker i's incentive constraint at p is always satisfied given zero effort e_i^p, every implementable profile must therefore satisfy every incentive constraint and the feasible set of the mixed-integer nonlinear program in Appendix B coincides with that of the convex program. □

A.4 Proof of Theorem 1

Assume that $V^p > V^q$ and that worker i is assigned to project q but capped at project p. The team-coordinated effort transfer in Section 5.1 offers a contradiction at the optimum. If $A^q \subset N$, I construct the following transfer:

$$x_j = (\delta(\mathbf{M}^{(A^q)})^{-1}\boldsymbol{\epsilon}_i)_j \text{ for } j \in A^q \text{ and } x_j = 0 \text{ for } j \notin A^q$$

The inverse $(\mathbf{M}^{(A^q)})^{-1}$ exists and is nonnegative because $\mathbf{M}$ is positive definite and has negative off-diagonal entries by Appendix A.5. Therefore $\mathbf{M}^{(A^q)}$ is a M-matrix because it inherits positive definiteness from $\mathbf{M}$ by Sylvester's criterion and has negative off-diagonal entries. M-matrices have nonnegative inverses.

Using this transfer, the manager moves an assigned worker's effort from q to p to precisely offset the total complementarity forgone at q and exploit the total complementarity gained at p. To show why this transfer works, examine how the slacks at project q change, which is where the naïve transfer failed. I use a block matrix computation. Let $\overline{A^q}$ be the set of all workers not assigned to project q. Partition $\mathbf{M}$ into blocks according to whether a worker is assigned to project q. To be clear, $\mathbf{M}^{(A^q),(\overline{A^q})}$ and $\mathbf{M}^{(\overline{A^q}),(A^q)}$ are submatrices of $\mathbf{M}$ containing complementarities $(-g_{ij})$ between workers assigned to q and those who are not. The change in slacks is

$$\Delta\mathbf{s}^q = \mathbf{M}\mathbf{x} = \begin{bmatrix} \mathbf{M}^{(A^q)} & \mathbf{M}^{(A^q),(\overline{A^q})} \\ \mathbf{M}^{(\overline{A^q}),(A^q)} & \mathbf{M}^{(\overline{A^q})} \end{bmatrix} \begin{bmatrix} (x_j)_{j\in A^q} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{M}^{(A^q)}(x_j)_{j\in A^q} \\ \mathbf{M}^{(\overline{A^q}),(A^q)}(x_j)_{j\in A^q} \end{bmatrix}$$

The first $|A^q|$ rows correspond to the incentive constraints of workers on the project team at q, and this is where the transfer $\mathbf{x}$ operates. By construction of x_j, we compute $\mathbf{M}^{(A^q)}(x_j)_{j\in A^q} = \delta\boldsymbol{\epsilon}_i$. Worker j's slack at q remains exactly the same as before the transfer. His decrease in effort at q matches the loss in complementarity driven by the reduction in his teammates' efforts, where all reductions are calibrated in the same way. Only worker i's slack at q changes; it rises by δ. The remaining rows correspond to workers not assigned to q. They lose some slack, but recall that under Assumption 2, incentive constraints do not bind at zero effort. Thus, they have strictly positive slack to absorb the loss given δ is sufficiently small. All of the same observations hold in reverse at project p.

Notice this covers all cases: capping worker i at project p means he can be either assigned or unassigned.

For the case when $V^p = V^q$, see Lemma 5. □

A.5 Proof of Lemma 4

By Appendix A.8, $\mathbf{M}$ is an M-matrix so $\mathbf{M}^{-1} \geq \mathbf{0}$.

Note that $\mathbf{M} = \mathbf{C} - \mathbf{R} - \mathbf{G}$ is a regular splitting because $\mathbf{G}$ has zero diagonal and positive definiteness gives $(\mathbf{C} - \mathbf{R})_{ii} = c_i - r_i > 0$. Hence $(\mathbf{C} - \mathbf{R})^{-1} \geq \mathbf{0}$ and $\mathbf{G} \geq \mathbf{0}$. Every regular splitting of a nonsingular M-matrix is convergent (Theorem 2.3 of Chapter 6 in Berman and Plemmons (1994)), and therefore the spectral radius $\rho((\mathbf{C} - \mathbf{R})^{-1}\mathbf{G}) < 1$. The Neumann series for $(\mathbf{I} - (\mathbf{C} - \mathbf{R})^{-1}\mathbf{G})^{-1}$ converges, and $\mathbf{M}^{-1} = (\mathbf{I} - (\mathbf{C} - \mathbf{R})^{-1}\mathbf{G})^{-1}(\mathbf{C} - \mathbf{R})^{-1} = \sum_{t=1}^{\infty}((\mathbf{C} - \mathbf{R})^{-1}\mathbf{G})^{t-1}(\mathbf{C} - \mathbf{R})^{-1}$. □

A.6 Proof of Lemma 5

Assume that $V^p = V^q$. The stationarity condition for the Lagrangian of the manager's optimization problem with respect to e_i^p is

$$V^p + \sum_{j \neq i} g_{ij}\lambda_j^p - c_i\lambda_i^p - r_i \sum_{q \neq p} \lambda_i^q + \mu_i^p = 0$$

where λ_i^p and μ_i^p are the multipliers on the (i, p) incentive and nonnegativity constraints. Written in terms of worker i's sum of multipliers on the incentive constraints $\Lambda_i := \sum_{p=1}^m \lambda_i^p$ is:

$$V^p + \textstyle\sum_{j \neq i} g_{ij}\lambda_j^p + \mu_i^p = (c_i - r_i)\lambda_i^p + r_i\Lambda_i$$

where $\Lambda_i := \sum_{p=1}^m \lambda_i^p$. Subtract the stationarity condition for (i, q) from (i, p). The multipliers satisfy the following equality

$$\textstyle\sum_{j \neq i} g_{ij}(\lambda_j^p - \lambda_j^q) + (\mu_i^p - \mu_i^q) = (c_i - r_i)(\lambda_j^p - \lambda_j^q).$$

Stack the equations for each worker

$$\boldsymbol{\mu}^p - \boldsymbol{\mu}^q = \mathbf{M}(\boldsymbol{\lambda}^p - \boldsymbol{\lambda}^q).$$

Suppose the two projects are staffed differently so that $\mathbf{d} := \boldsymbol{\lambda}^p - \boldsymbol{\lambda}^q \neq \mathbf{0}$. Because $\mathbf{M}$ is positive definite by Assumption 1 which I show in Appendix A.1, $\mathbf{d}^\intercal\mathbf{M}\mathbf{d} > \mathbf{0}$ while

the left hand side has $\mathbf{d}^\intercal(\boldsymbol{\mu}^p - \boldsymbol{\mu}^q) = -\boldsymbol{\lambda}^p\mu^q - \boldsymbol{\lambda}^q\mu^p \leq 0$ which is a contradiction. Thus it must be that $\boldsymbol{\lambda}^p = \boldsymbol{\lambda}^q$ and $\boldsymbol{\mu}^p = \boldsymbol{\mu}^q$.

For the second statement, fix a worker i capped at projects p and q and without loss of generality let $V^p \geq V^q$. Capping requires $\mu_i^p = 0$ and $\lambda_i^p = 0$, and likewise at q. Row i of the stationarity condition therefore loses its own-multiplier term $(c_i - r_i)\lambda_i^p$ and collapses to

$$V^p = r_i\Lambda_i - \sum_{j \neq i} g_{ij}\lambda_j^p$$

and identically at q. Subtracting the two, the term $r_i\Lambda_i$ cancels because it carries no project index:

$$V^p - V^q = \sum_{j \neq i} g_{ij}(\lambda_j^q - \lambda_j^p).$$

Here, I exploit a result on the monotonicity of λ that is established using linear complementarity theory.

Lemma 8. *If $V^p \geq V^q$, then $\boldsymbol{\lambda}^p \geq \boldsymbol{\lambda}^q$.*

Proof. There is a linear complementarity relationship between λ_i^p and μ_i^p.

1. If $\lambda_i^p > 0$, then $e_i^p > 0$ and $\mu_i^p = 0$.
2. If $\mu_i^p > 0$, then $e_i^p = 0$ and $\lambda_i^p = 0$.

Rearrange the stationarity condition for e_i^p:

$$\mu_i^p = (c_i - r_i)\lambda_i^p - \sum_{j \neq i} g_{ij}\lambda_j^p + r_i\Lambda_i - V^p \implies \boldsymbol{\mu}^p = \mathbf{M}\boldsymbol{\lambda}^p + \mathbf{R}\boldsymbol{\Lambda} - V^p\mathbf{1}_n$$

where $\mathbf{M} := \mathbf{C} - \mathbf{R} - \mathbf{G}$ and $\boldsymbol{\Lambda} = [\Lambda_1, ..., \Lambda_n]^\intercal$. Write $\mathbf{w}^p := \mathbf{R}\boldsymbol{\Lambda} - V^p\mathbf{1}_n$. The following are two facts established in linear complementarity theory.

1. Because $\mathbf{M}$ is positive definite, $\texttt{LCP}(\mathbf{M}, \mathbf{w}^p)$ has a unique solution for each right-hand vector $\mathbf{w}^p$ (Theorem 3.1.6 of Cottle et al. (2009)).
2. Because $\mathbf{M}$ is a Z-matrix, the unique solution $\boldsymbol{\lambda}^*(\mathbf{w}^p)$ to $\texttt{LCP}(\mathbf{M}, \mathbf{w}^p)$ is the least element of the feasible set (Theorem 3.11.6 of Cottle et al. (2009)).

Clearly $V^p \geq V^q$ implies $\mathbf{w}^p \leq \mathbf{w}^q$ (component-wise). Then $\mathbf{M}\boldsymbol{\lambda} + \mathbf{w}^q \geq \mathbf{M}\boldsymbol{\lambda} + \mathbf{w}^p$. This implies that $\{\boldsymbol{\lambda} \mid \mathbf{M}\boldsymbol{\lambda} + \mathbf{w}^p \geq \mathbf{0}\} \subseteq \{\boldsymbol{\lambda} \mid \mathbf{M}\boldsymbol{\lambda} + \mathbf{w}^q \geq \mathbf{0}\}$. By the least-element

property, the solution $\boldsymbol{\lambda}$ must satisfy:

$$\boldsymbol{\lambda}^q = \min_{\boldsymbol{\lambda} \geq \mathbf{0}} \{\boldsymbol{\lambda} \mid \mathbf{M}\boldsymbol{\lambda} + \mathbf{w}^q \geq \mathbf{0}\} \leq \min_{\boldsymbol{\lambda} \geq \mathbf{0}} \{\boldsymbol{\lambda} \mid \mathbf{M}\boldsymbol{\lambda} + \mathbf{w}^p \geq \mathbf{0}\} = \boldsymbol{\lambda}^p$$

This establishes monotonicity of $\boldsymbol{\lambda}^p$ with respect to V^p. □

Because $V^p \geq V^q$, we have $\boldsymbol{\lambda}^p \geq \boldsymbol{\lambda}^q$. As $g_{ij} \geq 0$, the right-hand side is nonpositive, so $V^p \leq V^q$ and therefore $V^p = V^q$. Because p and q were arbitrary projects at which worker i is capped, $V^{\phi_i+1} = \cdots = V^{\alpha_i}$. □

A.7 Proof of Proposition 1

The principal's objective is concave in $\mathbf{e}$ and every constraint is linear, so the KKT conditions are necessary and sufficient for optimality. Stationarity reads

$$\nabla_{\mathbf{e}} V(\mathbf{e}) = \mathbf{W}\boldsymbol{\lambda} - \boldsymbol{\mu}, \qquad \boldsymbol{\lambda}, \boldsymbol{\mu} \geq \mathbf{0},$$

with complementary slackness on the incentive and nonnegativity constraints.

($\Rightarrow$) Suppose every assigned worker is uncapped in the optimum $\mathbf{e}^*$, and let $A := \{(i,p) \mid e_i^{p*} > 0\}$. Every incentive constraint in A binds and $\mathbf{e}^*_{A^c} = \mathbf{0}$, so $\mathbf{W}_{(A)}\mathbf{e}^*_A = \boldsymbol{\beta}_A$. Because $\mathbf{W}_{(A)}$ is a principal submatrix of a positive definite matrix, by Sylvester's criterion it is positive definite and invertible, so $\mathbf{e}^* = \mathbf{e}(A)$ and $\mathbf{e}_A(A) > \mathbf{0}$. By **??**, incentive constraints are slack at zero effort, so $\boldsymbol{\lambda}_{A^c} = \mathbf{0}$. Moreover, $\boldsymbol{\mu}_A = \mathbf{0}$ because $\mathbf{e}^*_A > \mathbf{0}$. The A and A^c rows of stationarity become

$$\nabla_{\mathbf{e}_A} V(\mathbf{e}(A)) = \mathbf{W}_{(A)}\boldsymbol{\lambda}_A, \qquad \nabla_{\mathbf{e}_{A^c}} V(\mathbf{e}(A)) = \mathbf{W}_{A^cA}\boldsymbol{\lambda}_A - \boldsymbol{\mu}_{A^c},$$

so $\boldsymbol{\lambda}_A = \mathbf{W}^{-1}_{(A)} \nabla_{\mathbf{e}_A} V(\mathbf{e}(A))$ and $\boldsymbol{\mu}_{A^c} = \mathbf{W}_{A^cA}\boldsymbol{\lambda}_A - \nabla_{\mathbf{e}_{A^c}} V(\mathbf{e}(A))$. Every constraint indexed by A (incentive) and A^c (nonnegativity) binds, so again **??** gives $\boldsymbol{\lambda}_A > \mathbf{0}$ and $\boldsymbol{\mu}_{A^c} > \mathbf{0}$, which are conditions (2) and (3).

($\Leftarrow$) Suppose A satisfies conditions (1)-(3). By (1), $\mathbf{e}(A) \geq \mathbf{0}$ and every incentive constraint in A binds. Since $\mathbf{e}(A)$ satisfies the incentive constraint wherever effort is positive, Lemma 2 gives $\mathbf{E} \leq \tilde{\mathbf{E}}$, so by Assumption 2 every incentive constraint in A^c holds strictly at zero effort. Therefore, $\mathbf{e}(A)$ is feasible. Set $\boldsymbol{\lambda}_A$ and $\boldsymbol{\mu}_{A^c}$ as in (2) and (3), and $\boldsymbol{\lambda}_{A^c} = \mathbf{0}$, $\boldsymbol{\mu}_A = \mathbf{0}$. These multipliers are nonnegative, satisfy stationarity by

construction, and satisfy complementary slackness, so $\mathbf{e}(A)$ is optimal.

For uniqueness, let $\mathbf{e}'$ be any optimum. Optimal qualities are unique because V is strictly concave in $\mathbf{Q}$, so $\nabla_{\mathbf{e}} V(\mathbf{e}') = \nabla_{\mathbf{e}} V(\mathbf{e}(A))$, and the multipliers above also satisfy KKT at $\mathbf{e}'$. Complementary slackness with $\boldsymbol{\mu}_{A^c} > \mathbf{0}$ forces $\mathbf{e}'_{A^c} = \mathbf{0}$, and with $\boldsymbol{\lambda}_A > \mathbf{0}$ forces every incentive constraint in A to bind, so $\mathbf{W}_{(A)}\mathbf{e}'_A = \boldsymbol{\beta}_A$ and $\mathbf{e}' = \mathbf{e}(A)$. In the unique optimum, every worker is therefore uncapped on A and unassigned on A^c. □

A.8 Proof of Lemma 6

In $\hat{\mathbf{e}}$, the block corresponding to project p satisfies

$$(\mathbf{C} - \mathbf{R} - \mathbf{G})\hat{\mathbf{e}}^p + \mathbf{R}\hat{\mathbf{E}} = \boldsymbol{\beta}^p$$

where $\hat{\mathbf{E}} = [\sum_{p=1}^m \hat{e}_1^p \cdots \sum_{p=1}^m \hat{e}_n^p]$ is the total effort vector associated with $\hat{\mathbf{e}}$. Summing over all p, we get

$$\underbrace{((\mathbf{C} - \mathbf{R} - \mathbf{G}) + m\mathbf{R})}_{=\mathbf{P}} \hat{\mathbf{E}} = \sum_{p=1}^m \boldsymbol{\beta}^p$$

Therefore $\hat{\mathbf{E}} = \tilde{\mathbf{E}}$ and we have that

$$\hat{\mathbf{e}}^p = (\mathbf{C} - \mathbf{R} - \mathbf{G})^{-1}(\boldsymbol{\beta}^p - \mathbf{R}\tilde{\mathbf{E}})$$

I show in Appendix A.1 that Assumption 1 implies $\mathbf{M} := \mathbf{C} - \mathbf{R} - \mathbf{G}$ is a positive definite M-matrix as it is also symmetric with negative off-diagonals. Its inverse is nonnegative and has a strictly positive diagonal. Reading off each entry gives:

$$\hat{e}_i^p = \sum_{j=1}^n (\mathbf{M}^{-1})_{ij}(\beta_j^p - r_j\tilde{E}_j) \geq (\mathbf{M}^{-1})_{ii}(\beta_i^p - r_i\tilde{E}_i) > 0$$

where the last inequality holds by Assumption 2. □

A.9 Proof of Proposition 2

The KKT condition of the manager's problem, which is necessary and sufficient for optimality as all constraints are linear in efforts and the objective is concave, in vector

form, is:

$$\nabla_{\mathbf{e}} V(Q^1, ..., Q^m) + ((\mathbf{G} \otimes \mathbf{I}_m) - (\mathbf{C} \otimes \mathbf{I}_m + \mathbf{R} \otimes (\mathbf{1}_m \mathbf{1}_m^\intercal - \mathbf{I}_m)))\boldsymbol{\lambda} + \boldsymbol{\mu} = \mathbf{0}$$

where $\boldsymbol{\lambda} \geq \mathbf{0}$ and $\boldsymbol{\mu} \geq \mathbf{0}$ are the vectors of Lagrange multipliers for the incentive constraints and nonnegativity conditions. This is equivalent to $\nabla_{\mathbf{e}} V(Q^1, ..., Q^m) = \mathbf{W}\boldsymbol{\lambda} - \boldsymbol{\mu}$. If $(i, p) \in A^p$, then $\frac{\partial V}{\partial Q^p} = (\mathbf{W}\boldsymbol{\lambda})_i^p$. If $(i, p) \notin A^p$, then $\frac{\partial V}{\partial Q^p} \leq (\mathbf{W}\boldsymbol{\lambda})_i^p$. Keeping every worker assigned and uncapped at every project is optimal if and only if $\boldsymbol{\mu} = \mathbf{0}$ and $\nabla_{\mathbf{e}} V(\hat{\mathbf{e}}) = \mathbf{W}\boldsymbol{\lambda}$, which means $\mathbf{W}^{-1} \nabla_{\mathbf{e}} V(\hat{\mathbf{e}}) = \boldsymbol{\lambda} \geq \mathbf{0}$. □

A.10 Proof of Proposition 3

Let $(Q^{1*}, ..., Q^{m*})$ be the vector of total efforts that solves the manager's problem. It is unique because the feasible set is compact and the manager's valuation is strictly concave. Now, suppose by contradiction that $Q^{p*} < Q^{q*}$ for some $p < q$. Note that the effort profile that swaps efforts between projects p and q i.e. $Q^{p\prime} = Q^{q*}$ and $Q^{q\prime} = Q^{p*}$ is feasible because each worker's incentive constraint remains exactly as it was. Since the valuation is swap-increasing, $(Q^{1\prime}, ..., Q^{m\prime})$ gives a weakly higher value for the manager, which is a contradiction.

Now assume that $p < q$ and $Q^{p*} > Q^{q*}$. Optimality gives $\nabla V(Q^*) \cdot (Q - Q^*) \leq 0$ for every feasible total effort vector Q. Applying this to the swap Q' total effort profile gives $(V^p - V^q)(Q^{q*} - Q^{p*}) \leq 0$. If $Q^{p*} > Q^{q*}$, then it must be that $V^p \geq V^q$. If $Q^{p*} = Q^{q*}$, then the inequality is vacuously true; the inequality $V^p \geq V^q$ must hold because V is swap-increasing. □

A.11 Proof of Proposition 4

I first show that there always exists a worker-symmetric optimal project assignment. Note that the manager's objective is strictly concave over $(Q^1, ..., Q^m)$. Since the feasible set is convex, there exists a unique maximizer in total efforts per project $(Q^{1*}, ..., Q^{m*})$. Now, take any optimal project assignment $\{\{\hat{e}_i^p\}_{p=1}^m\}_{i=1}^n$ and consider the average assignment:

$$\bar{e}_i^p := \bar{e}^{p*} = \frac{1}{n} \sum_{j=1}^{n} \hat{e}_j^p$$

The total efforts at each project remain unchanged, and so $\{\{\bar{e}_i^p\}_{p=1}^m\}_{i=1}^n$ also achieves the manager's maximum. To show it is feasible, note that since $\{\{\hat{e}_i^p\}_{p=1}^m\}_{i=1}^n$ is feasible, then:

$$\beta + g\sum_{j\neq i}\hat{e}_j^p \geq c\hat{e}_i^p + r\sum_{q\neq p}\hat{e}_i^q \implies n\beta + g\sum_{j\neq i}\sum_{\ell=1}^n \hat{e}_\ell^p \geq c\sum_{j=1}\hat{e}_j^p + r\sum_{q\neq p}\sum_{j=1}^n \hat{e}_i^q$$

Now we need to show the incentive constraints for $\{\{\bar{e}_i^p\}_{p=1}^m\}_{i=1}^n$ also holds. Take any incentive constraint.

$$\begin{aligned}
\beta + g\sum_{j\neq i}\bar{e}_j^p - c\bar{e}_i^p - r\sum_{q\neq p}\bar{e}_i^q &= \beta + g\sum_{j\neq i}\left(\frac{1}{n}\sum_{\ell=1}^n \hat{e}_\ell^p\right) - c\left(\frac{1}{n}\sum_{j=1}^n \hat{e}_j^p\right) - r\sum_{q\neq p}\left(\frac{1}{n}\sum_{j=1}^n \hat{e}_j^q\right) \\
&= \beta + g\left(\frac{n-1}{n}\sum_{\ell=1}^n \hat{e}_\ell^p\right) - c\left(\frac{1}{n}\sum_{j=1}^n \hat{e}_j^p\right) - r\sum_{q\neq p}\left(\frac{1}{n}\sum_{j=1}^n \hat{e}_j^q\right) \\
&= \frac{1}{n}\left(n\beta + g\left((n-1)\sum_{\ell=1}^n \hat{e}_\ell^p\right) - c\left(\sum_{j=1}^n \hat{e}_j^p\right) - r\sum_{q\neq p}\left(\sum_{j=1}^n \hat{e}_j^q\right)\right) \\
&= \frac{1}{n}\left(n\beta + g\sum_{j\neq i}\sum_{\ell=1}^n \hat{e}_\ell^p - c\sum_{j=1}^n \hat{e}_j^p - r\sum_{q\neq p}\sum_{j=1}^n \hat{e}_j^q\right) \\
&\geq 0
\end{aligned}$$

(1) In the worker-symmetric optimal project assignment, every worker plays the same optimal effort at every project. Because the Cobb-Douglas objective satisfies the Inada conditions, every worker must play positive effort at every project.

(2) The worker-symmetric optimal project assignment solves the symmetric problem:

$$\begin{aligned}
&\max_{\{e^p\}_{p=1}^m} \sum_{p=1}^m \gamma^{(p)}\ln(e^p) \\
&\text{s.t. } \beta + g(n-1)e^p \geq ce^p + r\sum_{q\neq p} e^q
\end{aligned}$$

A unique maximizer $\{e^{p*}\}_{p=1}^m$ exists, and the KKT conditions are both necessary and sufficient. Consider project s and rewrite the incentive constraint in terms of the total effort in the entire system:

$$\beta + g(n-1)e^s \geq ce^s + r\sum_{p\neq s} e^p \implies \beta + (g(n-1) - c + r)e^s \geq r\sum_{p=1}^m e^p$$

For any projects s at which the incentive constraint binds, we have:

$$\beta + (g(n-1) - c + r)e^{s*} = r\sum_{p=1}^{m} e^{p*} \implies e^{s*} = \frac{\beta - r\sum_{p=1}^{m} e^{p*}}{c - r - g(n-1)}$$

For project t with a slack incentive constraint, we have

$$\beta + (g(n-1) - c + r)e^{t*} > r\sum_{q=1}^{m} e^{q*} \implies e^{t*} < \frac{\beta - r\sum_{q=1}^{m} e^{q*}}{c - r - g(n-1)}$$

This means $e^{t*} < e^{s*}$ for any project t and s with slack and binding incentive constraints, respectively.

All projects with binding incentive constraints have the same optimal effort, which depends on the total effort in the entire system. The first-order condition for any project s with a binding incentive constraint is

$$\frac{\gamma^{(s)}}{e^{s*}} = \lambda^{s*}(c - g(n-1) - r) + r\sum_{p=1}^{m} \lambda^{p*} \geq r\sum_{p=1}^{m} \lambda^{p*}$$

For project t with a slack incentive constraint ($\lambda^{t*} = 0$), its first-order condition is:

$$\frac{\gamma^{(t)}}{e^{t*}} + \lambda^{t*}(g(n-1) - c + r) - r\sum_{q=1}^{m} \lambda^{q*} = 0 \implies \frac{\gamma^{(t)}}{e^{t*}} = r\sum_{p=1}^{m} \lambda^{p*}$$

Linking both inequalities together yields:

$$\gamma^{(s)} \geq e^{s*}\left(r\sum_{p=1}^{m} \lambda^{q*}\right) > e^{t*}\left(r\sum_{q=1}^{m} \lambda^{q*}\right) = \gamma^{(t)}$$

Every project with a binding incentive constraint has a strictly larger γ-weight than every project with a slack incentive constraint. Since projects are ordered $\gamma^{(1)} > \cdots > \gamma^{(m)}$, that means there exists a project index ϕ^* that for all projects $s \leq \phi^*$, $\gamma^{(s)} \geq \gamma^{(\phi*)}$ with a binding incentive constraint and for all projects $t > \phi^*$, $\gamma^{(t)} < \gamma^{(\phi*)}$ with a slack incentive constraint. The formula for ϕ^* is derived in **(4)**.

(3) I first compute the optimal total effort in the entire system, which is a sufficient statistic for efforts at projects with a binding incentive constraint. All projects with binding incentive constraints have the same optimal effort. The following substitutes

in the first-order condition for projects with slack incentive constraints.

$$\sum_{p=1}^{m} e^{p*} = \sum_{p=1}^{\phi^*} e^{p*} + \sum_{t=\phi^*+1}^{m} e^{t*} = \phi^* e^{s*} + \left(\frac{1}{r\sum_{p=1}^{m}\lambda^{p*}}\right)\sum_{t=\phi^*+1}^{m}\gamma^{(t)}$$

From the binding incentive constraint on project s, we obtain:

$$\beta + (g(n-1) - c + r)e^{s*} = r\sum_{p=1}^{m} e^{p*} \implies \beta = (c - r - g(n-1) + r\phi^*)e^{s*} + \left(\frac{1}{\sum_{p=1}^{m}\lambda^{p*}}\right)\sum_{t=q^*+1}^{m}\gamma^{(t)}$$

Next, we determine a relationship between the multipliers $\sum_{p=1}^{m}\lambda^{p*}$ and the relative project weights γ. Summing the first-order conditions for all projects s with binding incentive constraints gives:

$$\frac{\gamma^{(s)}}{e^{s*}} = (c - r - g(n-1))\lambda^{s*} + r\sum_{p=1}^{m}\lambda^{p*} \implies \sum_{p=1}^{\phi^*}\frac{\gamma^{(p)}}{e^{p*}} = (c - r - g(n-1))\sum_{p=1}^{\phi^*}\lambda^{p*} + \phi^* r\sum_{p=1}^{m}\lambda^{p*}$$

Note that $\sum_{p=1}^{\phi^*}\lambda^{p*} = \sum_{p=1}^{m}\lambda^{p*}$ because $\lambda^{t*} = 0$ for all $t > \phi^*$. Together with the fact that all efforts at projects with binding incentive constraints are identical, we can write:

$$\frac{1}{e^{s*}}\sum_{p=1}^{\phi^*}\gamma^{(p)} = (c - r - g(n-1)) + \phi^* r)\sum_{p=1}^{m}\lambda^{p*} \implies e^{s*}(c - r - g(n-1)) + \phi^* r) = \frac{\sum_{p=1}^{\phi^*}\gamma^{(p)}}{\sum_{p=1}^{m}\lambda^{p*}}$$

Finally, substitute this into the equation for β:

$$\beta = \frac{\sum_{p=1}^{\phi^*}\gamma^{(p)}}{\sum_{p=1}^{m}\lambda^{p*}} + \left(\frac{1}{\sum_{p=1}^{m}\lambda^{p*}}\right)\sum_{t=\phi^*+1}^{m}\gamma^{(t)} \implies \sum_{p=1}^{m}\lambda^{p*} = \frac{\sum_{p=1}^{m}\gamma^{(p)}}{\beta}$$

Now we can directly compute e^{s*} and e^{t*}:

$$e^{s*} = \frac{\beta - \left(\frac{1}{\sum_{p=1}^{m}\lambda^{p*}}\right)\sum_{t=\phi^*+1}^{m}\gamma^{(t)}}{c - r - g(n-1) + r\phi^*} \implies e^{s*} = \frac{\beta}{c - r - g(n-1) + \phi^* r}\left(\frac{\sum_{p=1}^{\phi^*}\gamma^{(p)}}{\sum_{p=1}^{m}\gamma^{(p)}}\right)$$

$$e^{t*} = \frac{\gamma^{(t)}}{r\sum_{p=1}^{m}\lambda^{p*}} = \frac{\beta}{r}\left(\frac{\gamma^{(t)}}{\sum_{p=1}^{m}\gamma^{(p)}}\right)$$

For project s with a binding incentive constraint, the first-order condition inequality with the closed form for e^{s*} gives:

$$\frac{\gamma^{(s)}}{e^{s*}} \geq r \sum_{p=1}^{m} \lambda^{p*} \implies \gamma^{(s)} \left(\frac{c - r - g(n-1) + \phi^* r}{\beta} \right) \left(\frac{\sum_{p=1}^{m} \gamma^{(p)}}{\sum_{p=1}^{\phi^*} \gamma^{(p)}} \right) \geq r \left(\frac{\sum_{p=1}^{m} \gamma^{(p)}}{\beta} \right)$$

The right-hand inequality simplifies to

$$\frac{\gamma^{(s)}}{r} \geq \frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{c - r - g(n-1) + \phi^* r}$$

Since this inequality holds for all $s \leq \phi^*$, it must be that

$$\frac{\gamma^{\phi^*}}{r} \geq \frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{c - r - g(n-1) + \phi^* r}$$

Note that project $\phi^* + 1$ must have a slack incentive constraint. We can substitute the formulas for e^{q*} and e^{q*+1} into the inequality $e^{\phi^*+1} < e^{q*}$:

$$\frac{\beta}{r} \left(\frac{\gamma^{(\phi^*+1)}}{\sum_{p=1}^{m} \gamma^{(p)}} \right) < \frac{\beta}{c - r - g(n-1) + \phi^* r} \left(\frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{\sum_{p=1}^{m} \gamma^{(p)}} \right) \implies \frac{\gamma^{\phi^*+1}}{r} < \frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{c - r - g(n-1) + \phi^* r}$$

Thus ϕ^* is a natural number that satisfies:

$$\frac{\gamma^{(\phi^*+1)}}{r} < \frac{\sum_{p=1}^{\phi^*} \gamma^{(p)}}{c - r - g(n-1) + \phi^* r} \leq \frac{\gamma^{(\phi^*)}}{r}$$

If there was another natural number $\hat{q} \neq \phi^*$ that satisfies the above thresholds, then $\hat{q}$ induce $\hat{e}^s \neq e^{s*}$ and $\hat{e}^t \neq e^{t*}$ for all projects with binding and slack incentive constraints, respectively. However, this contradicts the uniqueness of the optimal e^*. From the above, we can immediately see that ϕ^* is the maximum project index that satisfies

$$\frac{\sum_{p=1}^{\phi} \gamma^{(\phi)}}{c - r - g(n-1) + \phi r} \leq \frac{\gamma^{(\phi)}}{r} \iff \frac{\sum_{p=1}^{\phi} \gamma^{(p)}}{\gamma^{(\phi)}} - \phi \leq \frac{c - r - g(n-1)}{r}.$$

(4) I show that ϕ^* is non-increasing in r using the established bounds about ϕ^*. From the right inequality, we have the equivalent statement

$$\frac{\sum_{p=1}^{\phi^*}\gamma^{(p)}}{c-r-g(n-1)+\phi^* r} \leq \frac{\gamma^{(\phi^*)}}{r} \iff r\left(\sum_{p=1}^{\phi^*}\gamma^{(p)} - (\phi^*-1)\gamma^{(\phi^*)}\right) \leq \gamma^{(\phi^*)}(c-g(n-1))$$

Because we ordered $\gamma^{(1)} > \cdots > \gamma^{(m)}$,

$$\sum_{p=1}^{\phi^*}\gamma^{(p)} - \phi^*\gamma^{(\phi^*)} > 0 \implies \sum_{p=1}^{\phi^*}\gamma^{(p)} - (\phi^*-1)\gamma^{(\phi^*)} > 0$$

We derive the following upper bound on r in terms of ϕ^*:

$$r \leq \frac{\gamma^{(\phi^*)}(c-g(n-1))}{\sum_{p=1}^{\phi^*}\gamma^{(p)} - (\phi^*-1)\gamma^{(\phi^*)}} := U_{\phi^*}$$

From the left inequality that ϕ^* satisfies in (4a), we have the equivalent statement:

$$\frac{\gamma^{(\phi^*+1)}}{r} < \frac{\sum_{p=1}^{\phi^*}\gamma^{(p)}}{c-r-g(n-1)+\phi^* r} \iff r\left(\sum_{p=1}^{\phi^*}\gamma^{(p)} - (\phi^*-1)\gamma^{(\phi^*+1)}\right) > \gamma^{(\phi^*+1)}(c-g(n-1))$$

This establishes a lower bound on r in terms of ϕ^*:

$$r > \frac{\gamma^{(\phi^*+1)}(c-g(n-1))}{\sum_{p=1}^{\phi^*}\gamma^{(p)} - (\phi^*-1)\gamma^{(\phi^*+1)}} := L_{\phi^*}$$

Note that $U_{q+1} = L_q$ and U_q is strictly decreasing in q. I show this by computing the discrete difference

$$U_{q+1} - U_q = \frac{(\gamma^{(q+1)} - \gamma^{(q)})(c-g(n-1))\sum_{p=1}^{q}\gamma^{(p)}}{(\sum_{p=1}^{q}\gamma^{(p)} - (q-1)\gamma^{(q)})(\sum_{p=1}^{q}\gamma^{(p)} - (q-1)\gamma^{(q+1)})} < 0$$

Thus we have partitioned the r-space into m disjoint intervals:

$$(L_m, U_m] \cup (L_{m-1}, U_{m-1}] \cup \cdots \cup (L_1, U_1]$$

where $L_m = 0$ and $U_1 = c - g(n-1)$. For any r that satisfies $r \in [0, c-g(n-1))$, there exists an unique ϕ^* such that $L_{\phi^*} < r \leq U_{\phi^*}$. Thus ϕ^* is a non-increasing step

function of r given by:

$$\phi^*(r) = \max\{q \mid r \leq U_q\}$$

The bounds U_q are increasing in c and decreasing in g and n. Therefore, ϕ^* is a non-decreasing step function in c and a non-increasing step function in g and n. Because U_q is independent of β, then ϕ^* is also independent of β. □

A.12 Proof of Theorem 2

Rank projects by their marginal value in the optimum and apply Theorem 1 to conclude there exists a worker-specific project index α_i such that $e_i^p > 0$ if and only if $p \leq \alpha_i$. To relate α_i to χ_i, I solve for λ_i^p and μ_i^p in the following case analysis. The stationarity equation of the Lagrangian is:

$$V^p + g\sum_{j\neq i}\lambda_j^p - c_i\lambda_i^p - r_i\sum_{q\neq p}\lambda_i^q + \mu_i^p = 0$$

Written in terms of $\Lambda_i := \sum_{p=1}^m \lambda_i^p$ and $\Lambda^p := \sum_{i=1}^n \lambda_i^p$, we have:

$$V^p - \lambda_i^p(c_i + g - r_i) - r_i\Lambda_i + g\Lambda^p + \mu_i^p = 0$$

Case #1.1: $e_i^p > 0$.

Then $\mu_i^p = 0$. From the linear complementarity problem, we have

$$0 = \lambda_i^p(c_i + g - r_i) + r_i\Lambda_i - V^p - g\Lambda^p \implies \lambda_i^p = \frac{V^p + g\Lambda^p - r_i\Lambda_i}{c_i + g - r_i}$$

Since $\lambda_i^p \geq 0$, it must be that $V^p + g\Lambda^p \geq r_i\Lambda_i$.

Case #1.2: $e_i^p = 0$.

Then $\lambda_i^p = 0$ by the generic strict slackness assumption. We have

$$\mu_i^p = r_i\Lambda_i - V^p - g\Lambda^p$$

Putting both cases together, we have the following expressions for λ_i^p and μ_i^p:

$$\lambda_i^p = \frac{\max(V^p + g\Lambda^p - r_i\Lambda_i, 0)}{c_i + g - r_i} \qquad \mu_i^p = \max(r_i\Lambda_i - V^p - g\Lambda^p, 0)$$

If $V^p + g\Lambda^p > r_i\Lambda_i$, then $\lambda_i^p > 0$. By **??**, $\lambda_i^p > 0$ implies $e_i^p > 0$. If $V^p + g\Lambda^p < r_i\Lambda_i$, then $\mu_i^p > 0$, forcing $e_i^p = 0$. Finally, if $V^p + g\Lambda^p = r_i\Lambda_i$, then both $\lambda_i^p = \mu_i^p = 0$. This allows us to write $\alpha_i = \max\{p \mid V^p + g\Lambda^p \geq r_i\Lambda_i\}$ and also shows that $V^p + g\Lambda^p$ is non-increasing in p.

Note that the only worker-specific term that determines either multiplier is $r_i\Lambda_i$. I proceed with showing that $\chi_i \geq \chi_j$ implies $r_i\Lambda_i \geq r_j\Lambda_j$. Suppose by contradiction that $r_i\Lambda_i < r_j\Lambda_j$, then $r_j\Lambda_j > 0$ and $\Lambda_j > 0$ necessarily and there is at least one project with $\lambda_j^p > 0$ otherwise $\Lambda_j = 0$. Partition the project space into three sets: $X := \{p \mid V^p + g\Lambda^p > r_j\Lambda_j\}$, $Y := \{p \mid r_j\Lambda_j \geq V^p + g\Lambda^p > r_i\Lambda_i\}$ and $Z := \{p \mid V^p + g\Lambda^p \leq r_i\Lambda_i\}$.

On X, both λ_i^p and λ_j^p are strictly positive and because $r_i\Lambda_i < r_j\Lambda_j$, we have

$$\lambda_i^p = \frac{V^p + g\Lambda^p - r_i\Lambda_i}{c_i + g - r_i} > \frac{V^p + g\Lambda^p - r_j\Lambda_j}{c_i + g - r_i} = \left(\frac{c_j + g - r_j}{c_i + g - r_i}\right)\lambda_j^p$$

On Y, $\lambda_i^p > 0 = \lambda_j^p$ and on Z, both multipliers are zero. Therefore, summing both sides over all projects gives

$$\Lambda_i > \left(\frac{c_j + g - r_j}{c_i + g - r_i}\right)\Lambda_j$$

Multiply both sides by $r_i(c_i + g - r_i)$ and apply $\chi_i \geq \chi_j \iff r_i(c_j + g - r_j) \geq r_j(c_i + g - r_i)$ to obtain

$$r_i(c_i + g - r_i)\Lambda_i > r_i(c_j + g - r_j)\Lambda_j \geq r_j(c_i + g - r_i)\Lambda_j$$

This also holds iff

$$r_i(c_i + g)\Lambda_i > r_j(c_i + g)\Lambda_j$$

This implies $r_i\Lambda_i \geq r_j\Lambda_j$ which is a direct contradiction.

Because $V^p + g\Lambda^p$ is non-increasing in p, the active set for worker i is $\{p \mid V^p + g\Lambda^p \geq r_i\Lambda_i\}$. Thereby $\chi_i \geq \chi_j \implies r_i\Lambda_i \geq r_j\Lambda_j \implies \{p \mid V^p + g\Lambda^p \geq r_i\Lambda_i\} \subseteq \{p \mid V^p + g\Lambda^p \geq r_j\Lambda_j\} \implies \alpha_i \leq \alpha_j$ and $\phi_i \leq \phi_j$. □

A.13 Proof of Theorem 3

The stationarity equation of the Lagrangian is given by

$$V^p + \sum_{j \neq i} g_{ij} \lambda_j^p - c\lambda_i^p - r \sum_{q \neq p} \lambda_i^p + \mu_i^p = 0$$

If every worker is assigned to every project, then $\mu_i^p = 0$ for all (i, p). Add and subtract $r_i \lambda_i^p$ and stack the stationarity equation for the given project p over all workers and rearrange to obtain:

$$V^p \mathbf{1}_n = \mathbf{R}\boldsymbol{\Lambda} + \mathbf{M}\boldsymbol{\lambda}^p$$

Now sum over all projects:

$$(\textstyle\sum_{p=1}^m V^p)\mathbf{1}_n = \underbrace{(\mathbf{M} + m\mathbf{R})}_{=\mathbf{P}} \boldsymbol{\Lambda} \implies \boldsymbol{\Lambda} = \mathbf{P}^{-1}(\textstyle\sum_{p=1}^m V^p)\mathbf{1}_n$$

Substituting into the stationarity condition at project p, we have that

$$V^p \mathbf{1}_n = \mathbf{R}(\mathbf{P}^{-1}(\textstyle\sum_{p=1}^m V^p)\mathbf{1}_n) + \mathbf{M}\boldsymbol{\lambda}^p \implies \boldsymbol{\lambda}^p = \mathbf{M}^{-1}(V^p \mathbf{1}_n - \mathbf{R}(\mathbf{P}^{-1}(\textstyle\sum_{p=1}^m V^p)\mathbf{1}_n))$$

Distributing through:

$$\boldsymbol{\lambda}^p = \mathbf{M}^{-1} V^p \mathbf{1}_n - \mathbf{M}^{-1}\mathbf{R}\mathbf{P}^{-1}(\textstyle\sum_{p=1}^m V^p)\mathbf{1}_n \implies \boldsymbol{\lambda}^p = \mathbf{M}^{-1} V^p \mathbf{1}_n - \frac{1}{m}(\mathbf{M}^{-1} - \mathbf{P}^{-1})(\textstyle\sum_{p=1}^m V^p)\mathbf{1}_n$$

The implication uses the following matrix identity

$$\mathbf{M}^{-1}\mathbf{R}\mathbf{P}^{-1} = \tfrac{1}{m}\mathbf{M}^{-1}(\mathbf{P} - \mathbf{M})\mathbf{P}^{-1} = \tfrac{1}{m}(\mathbf{M}^{-1}\mathbf{P}\mathbf{P}^{-1} - \mathbf{M}^{-1}\mathbf{M}\mathbf{P}^{-1}) = \tfrac{1}{m}(\mathbf{M}^{-1} - \mathbf{P}^{-1})$$

Reading off worker i's row gives

$$\lambda_i^p = V^p(\mathbf{M}^{-1}\mathbf{1}_n)_i - \frac{1}{m}\sum_{p=1}^m V^p((\mathbf{M}^{-1}\mathbf{1}_n)_i - (\mathbf{P}^{-1}\mathbf{1}_n)_i)$$

Factoring out $(\mathbf{M}^{-1}\mathbf{1}_n)_i$, we obtain

$$\lambda_i^p = (\mathbf{M}^{-1}\mathbf{1}_n)_i (V^p - \underbrace{\tfrac{1}{m}\textstyle\sum_{p=1}^m V^p (1 - \frac{(\mathbf{P}^{-1}\mathbf{1}_n)_i}{(\mathbf{M}^{-1}\mathbf{1}_n)_i})}_{:=\delta_i})$$

Worker i is left uncapped at project p if and only if $\lambda_i^p > 0$. If $\rho_i > \rho_j$, then $\delta_i < \delta_j$. This implies $\{p \mid V^p > \delta_j\} \subseteq \{p \mid V^p > \delta_i\} \implies \phi_i \geq \phi_j$. □

A.14 Proof of Theorem 4

At the optimum, $(\mathbf{e}, \boldsymbol{\lambda}_\Phi)$ satisfy the KKT conditions:

$$\begin{aligned}\nabla_{\mathbf{e}} V(\mathbf{e}) - \mathbf{W}_\Phi^\intercal \boldsymbol{\lambda}_\Phi &= \mathbf{0} \\ \boldsymbol{\beta}_\Phi - \mathbf{W}_\Phi \mathbf{e} &= \mathbf{0}\end{aligned}$$

Differentiating with respect to parameter θ gives

$$\nabla^2_{\mathbf{e},\theta} V(\mathbf{e}) + \nabla^2_{\mathbf{e}} V(\mathbf{e}) \frac{\partial \mathbf{e}}{\partial \theta} - \mathbf{W}_\Phi^\intercal \frac{\partial \boldsymbol{\lambda}_\Phi}{\partial \theta} - \frac{\partial \mathbf{W}_\Phi^\intercal}{\partial \theta} \boldsymbol{\lambda}_\Phi = \mathbf{0}$$

$$\frac{\partial \boldsymbol{\beta}_\Phi}{\partial \theta} - \mathbf{W}_\Phi \frac{\partial \mathbf{e}}{\partial \theta} - \frac{\partial \mathbf{W}_\Phi}{\partial \theta} \mathbf{e} = \mathbf{0}$$

Thus we can write in matrix form

$$\underbrace{\begin{bmatrix} \nabla^2_{\mathbf{e}} V(\mathbf{e}) & -\mathbf{W}_\Phi^\intercal \\ -\mathbf{W}_\Phi & \mathbf{0} \end{bmatrix}}_{:=\mathbf{J}} \begin{bmatrix} \frac{\partial \mathbf{e}}{\partial \theta} \\ \frac{\partial \boldsymbol{\lambda}_\Phi}{\partial \theta} \end{bmatrix} = \begin{bmatrix} -\nabla^2_{\mathbf{e},\theta} V(\mathbf{e}) + \frac{\partial \mathbf{W}_\Phi^\intercal}{\partial \theta} \boldsymbol{\lambda}_\Phi \\ -\frac{\partial \boldsymbol{\beta}_\Phi}{\partial \theta} + \frac{\partial \mathbf{W}_\Phi}{\partial \theta} \mathbf{e} \end{bmatrix}$$

Note that $-\nabla^2_{\mathbf{e},\theta} V(\mathbf{e}) = \mathbf{0}$. I proceed to show that $\mathbf{J}$ is invertible. Suppose $\mathbf{x} \in \mathbb{R}^{mn}$ and $\mathbf{y} \in \mathbb{R}^F$ where $(\mathbf{x}, \mathbf{y}) \in \ker(\mathbf{J})$. Therefore:

$$\begin{aligned}\nabla^2_{\mathbf{e}} V(\mathbf{e}) \mathbf{x} - \mathbf{W}_F^\intercal \mathbf{y} &= \mathbf{0} \\ -\mathbf{W}_F \mathbf{x} &= \mathbf{0}\end{aligned}$$

Premultiplying the first equation by $\mathbf{x}^\intercal$ gives

$$\mathbf{x}^\intercal \nabla^2_{\mathbf{e}} V(\mathbf{e}) \mathbf{x} - \mathbf{x}^\intercal \mathbf{W}_F^\intercal \mathbf{y} = 0 \implies \mathbf{x}^\intercal \nabla^2_{\mathbf{e}} V(\mathbf{e}) \mathbf{x} = \mathbf{0}$$

By assumption that $\nabla^2_{\mathbf{e}} V(\mathbf{e})$ is strictly negative definite on $\ker(\mathbf{W}_\Phi^\intercal)$, it must be that $\mathbf{x} = \mathbf{0}$. Substituting back into the first equation, we get

$$\mathbf{W}_\Phi^\intercal \mathbf{y} = 0$$

Recall that $\mathbf{W}$ is invertible, so $\mathbf{W}_\Phi$ has full row rank and it must be that $\mathbf{y} = \mathbf{0}$. This concludes the argument that $\mathbf{J}$ is invertible. Therefore the Implicit Function Theorem applies, and

$$\begin{bmatrix} \frac{\partial \mathbf{e}}{\partial \theta} \\ \frac{\partial \boldsymbol{\lambda}_\Phi}{\partial \theta} \end{bmatrix} = \mathbf{J}^{-1} \begin{bmatrix} \frac{\partial \mathbf{W}_\Phi^\intercal}{\partial \theta} \boldsymbol{\lambda}_\Phi \\ \frac{\partial \boldsymbol{\beta}_\Phi}{\partial \theta} - \frac{\partial \mathbf{W}_\Phi}{\partial \theta} \mathbf{e} \end{bmatrix}$$

which implies that

$$\frac{\partial \mathbf{e}}{\partial \theta} = \mathbf{S}\left(\frac{\partial \mathbf{W}_\Phi^\intercal}{\partial \theta} \boldsymbol{\lambda}_\Phi\right) + \mathbf{Y}\left(\frac{\partial \boldsymbol{\beta}_\Phi}{\partial \theta} - \frac{\partial \mathbf{W}_\Phi}{\partial \theta} \mathbf{e}\right)$$

□

A.15 Proof of Proposition 5

Since every worker assigned to a project is left uncapped, $\mathbf{W}_\Phi$ is a square matrix. Moreover, it is invertible because it is a principal submatrix of $\mathbf{W}$ which is positive definite. Because $\mathbf{J}$ is invertible, the bottom-left block of $\mathbf{J}\mathbf{J}^{-1} = \mathbf{I}$, which requires $-\mathbf{W}_\Phi \mathbf{S} = \mathbf{0}$. By invertibility of $\mathbf{W}_\Phi$, we must have $\mathbf{S} = \mathbf{0}$. □

A.16 Proof of Proposition 6

When every worker is assigned and uncapped at every project, the optimal project assignment has, for each worker i and project p, $e_i^{p*} > 0$, and the incentive constraint binds

$$\beta_i^p + \sum_{j \neq i} g_{ji} e_j^{p*} = c_i e_i^{p*} + r_i \sum_{q \neq p} e_i^{q*}$$

Therefore, we have

$$\boldsymbol{\beta}^p = \mathbf{M}\mathbf{e}^p + \mathbf{R}\mathbf{E}\mathbf{1}_m$$

Now sum over all projects to obtain

$$\sum_{p=1}^m \boldsymbol{\beta}^p = \sum_{p=1}^m \left(\mathbf{M}\mathbf{e}^p + \mathbf{R}\mathbf{E}\mathbf{1}_m\right) \implies \sum_{p=1}^m \boldsymbol{\beta}^p = \mathbf{M}\mathbf{E}\mathbf{1}_m + m\mathbf{R}\mathbf{E}\mathbf{1}_m$$

Therefore the total effort vector satisfies $\mathbf{E}\mathbf{1}_m = \mathbf{P}^{-1}\sum_{p=1}^m \boldsymbol{\beta}^p$. Substituting into the incentive constraint for project p gives

$$\boldsymbol{\beta}^p = \mathbf{M}\mathbf{e}^p + \mathbf{R}\mathbf{P}^{-1}\sum_{p=1}^m \boldsymbol{\beta}^p \mathbf{1}_m \implies \mathbf{e}^p = \mathbf{M}^{-1}\boldsymbol{\beta}^p - \mathbf{M}^{-1}\mathbf{R}\mathbf{P}^{-1}\sum_{p=1}^m \boldsymbol{\beta}^p$$

Now we can differentiate

$$\frac{\partial \mathbf{e}^p}{\partial \beta_j^q} = \mathbf{M}^{-1}\boldsymbol{\epsilon}_j\delta_{pq} - \mathbf{M}^{-1}\mathbf{R}\mathbf{P}^{-1}\boldsymbol{\epsilon}_j$$

Reading off the i-th entry gives the formula in the theorem. All of the other equations are derived similarly.

To show the sign of $\frac{\partial e_i^{p*}}{\partial \beta_j^q}$, note that its matrix form is given by:

$$\frac{\partial \mathbf{e}^p}{\partial \boldsymbol{\beta}^q} = \mathbf{M}^{-1}\delta_{pq} - \mathbf{M}^{-1}\mathbf{R}\mathbf{P}^{-1} = \mathbf{M}^{-1}\delta_{pq} - \frac{1}{m}(\mathbf{M}^{-1} - \mathbf{P}^{-1})$$

The second equality hold because $\mathbf{R} = \frac{1}{m}(\mathbf{P} - \mathbf{M})$. Therefore:

$$\frac{\partial \mathbf{e}^p}{\partial \boldsymbol{\beta}^q} = \begin{cases} \frac{m-1}{m}\mathbf{M}^{-1} + \frac{1}{m}\mathbf{P}^{-1} & p = q \\ -\frac{1}{m}(\mathbf{M}^{-1} - \mathbf{P}^{-1}) & p \neq q \end{cases}$$

Therefore $\frac{\partial e_i^{p*}}{\partial \beta_j^p} > 0$ for all $j \neq i$ and $\frac{\partial e_i^{p*}}{\partial \beta_j^q} < 0$ for all $q \neq p$. □

A.17 Proof of Proposition 7

As an abuse of notation, $\mathbf{e}^p$ and $\boldsymbol{\beta}^p$ both refer to the efforts and preferences of workers restricted to those in A^p. Define $\mathbf{E} = [\mathbf{e}^1 \cdots \mathbf{e}_n]$. In an optimum with partial assignment, the binding incentive constraint for worker i at project p is

$$\beta_i^p + \sum_{j\neq i, j\in A^p} g_{ji} e_j^{p*} = c_i e_i^{p*} + r_i \sum_{q\neq p, q\in P_i} e_i^{q*}$$

In vector form, we have

$$\boldsymbol{\beta}^p = \mathbf{M}^p\mathbf{e}^p + (\mathbf{R}\mathbf{E}\mathbf{1}_m)^p$$

where $(\mathbf{RE1}_m)^p$ is the vector of total efforts for workers in A^p multiplied by r_i. Solving for $\mathbf{e}^p$ gives

$$\mathbf{e}^p = (\mathbf{M}^p)^{-1}\boldsymbol{\beta}^p - (\mathbf{M}^p)^{-1}(\mathbf{RE1}_m)^p$$

$\mathbf{M}^p$ is invertible because principal submatrices of positive definite matrices are also positive definite by Sylvester's Criterion. Now I solve for a worker's total effort

$$Q_i = \sum_{p \in P_i} e_i^p = \sum_{p \in P_i} \left(\sum_{j \in A^p} (\mathbf{M}^p)_{ij}^{-1} \beta_j^p - (\mathbf{M}^p)_{ij}^{-1} r_j Q_j \right)$$

Rearranging, I obtain:

$$Q_i + \sum_{j=1}^{n} \left(r_j \sum_{p \in P_i \cap P_j} (\mathbf{M}^p)_{ij}^{-1} \right) Q_j = \sum_{p \in P_i} \sum_{j \in A^p} (\mathbf{M}^p)_{ij}^{-1} \beta_j^p$$

Define $\hat{\beta}_i := \sum_{p \in P_i} \sum_{j \in A^p} (\mathbf{M}^p)_{ij}^{-1} \beta_j^p$ and $n \times n$ matrix $\mathcal{P}$ with the following entries:

$$\mathcal{P}_{ij} = \delta_{ij} + r_j \sum_{p \in P_i \cap P_j} (\mathbf{M}^p)_{ij}^{-1}$$

Given $\mathcal{P}$, we have the following linear system in workers' total efforts

$$\mathcal{P}(\mathbf{E1}_m) = \hat{\boldsymbol{\beta}}$$

Because V is strictly concave in total efforts, the optimal total effort is unique and $\mathcal{P}$ is invertible. To see the latter, let $\mathbf{K}$ be the sum of the inverses $(\mathbf{M}^{A^p})^{-1}$, each embedded in worker space with zeros for unassigned workers. Each matrix in the sum is positive semidefinite, so $\mathbf{K}$ is positive semidefinite. By definition, $\mathcal{P} = \mathbf{I} + \mathbf{KR} = (\mathbf{R}^{-1} + \mathbf{K})\mathbf{R}$. Since $\mathbf{R}^{-1} + \mathbf{K}$ is positive definite and $\mathbf{R}$ is invertible, $\mathcal{P}$ is invertible. Now, we have

$$\mathbf{E1}_m = \mathcal{P}^{-1}\hat{\boldsymbol{\beta}} \implies \frac{\partial(\mathbf{E1}_m)}{\partial \zeta} = \mathcal{P}^{-1}\boldsymbol{\xi}^{(\zeta)}$$

where $\boldsymbol{\xi}^{(\zeta)}$ is defined for each model parameter ζ as

$$\boldsymbol{\xi}_i^{(\beta_j^q)} = \begin{cases} (\mathbf{M}^q)_{ij}^{-1} & \text{if } i \in A^q \\ 0 & \text{o.w.} \end{cases} \quad ; \quad \boldsymbol{\xi}_i^{(c_j)} = -\sum_{p \in A_i \cap A_j} (\mathbf{M}^p)_{ij}^{-1} e_j^{p*}$$

$$\xi_i^{(r_j)} = \sum_{p \in A_i \cap A_j} (\mathbf{M}^p)^{-1}_{ij} (e_j^{p*} - Q_j^*) \;\; ; \;\; \xi_i^{(g_{j\ell})} = \sum_{p \in A_i \cap A_j \cap A_\ell} (\mathbf{M}^p)^{-1}_{ij} e_\ell^{p*} + (\mathbf{M}^p)^{-1}_{i\ell} e_j^{p*}$$

Now we can finally compute:

$$\frac{\partial \mathbf{e}^p}{\partial \zeta} = \frac{\partial}{\partial \zeta}\left((\mathbf{M}^p)^{-1}\boldsymbol{\beta}^p\right) - \frac{\partial}{\partial \zeta}\left((\mathbf{M}^p)^{-1}(\mathbf{RE1}_m)^p\right)$$

Each of the results follows from direct computation of the above equation and substituting in $\dfrac{\partial(\mathbf{E1}_m)}{\partial \zeta} = \mathcal{P}^{-1}\boldsymbol{\xi}^{(\zeta)}$. □

A.18 Proof of Proposition 8

Assume that $\mathbf{e}^*$ is an effort profile where worker i is capped at p so that $\frac{\partial U_i}{\partial e_i^p} > 0$ at $\mathbf{e}^*$. Let $\boldsymbol{\epsilon}_p$ be the p-th unit vector. For sufficiently small $t > 0$, $U_i(\mathbf{e}_i^* + t\boldsymbol{\epsilon}_p, \mathbf{e}_{-i}^*) > U_i(\mathbf{e}_i^*, \mathbf{e}_{-i}^*)$. Monotonicity of w_i gives $w_i(\mathbf{e}_i^* + t\boldsymbol{\epsilon}_p, \mathbf{e}_{-i}^*) \geq w_i(\mathbf{e}_i^*, \mathbf{e}_{-i}^*)$, and so we have constructed a profitable deviation for worker i. □

B The General MINLP

The manager's general problem is written as a mixed-integer nonlinear program (MINLP).

$$\begin{aligned}
&\max_{\mathbf{e},\mathbf{z}} \; V(Q^1, ..., Q^m) \\
&\text{s.t. } \mathbf{e} \geq \mathbf{0}, \\
&\mathbf{z} \in \{0,1\}^{mn}, \\
&\forall (i,p), \; e_i^p \leq M z_i^p \\
&\forall (i,p), \; \beta_i^p + \sum_{j \neq i} g_{ij} e_j^p - c_i e_i^p - r_i \sum_{q \neq p} e_i^q \geq -M(1 - z_i^p) \\
&\forall p, \; \sum_{i=1}^n e_i^p = Q^p
\end{aligned} \tag{P3}$$

where M is a sufficiently large positive constant.[18] z_i^p is a binary variable that acts as a switch to ensure that if $e_i^p > 0$, then the incentive constraint has to be satisfied. If

[18] Instead of using a Big-M constant, we can also write the manager's problem with disjunctive constraints: if $z_i^p = 1$, then $\beta_i^p + \sum_{j \neq i} g_{ij} e_j^p - c_i e_i^p - r_i \sum_{q \neq p} e_i^q \geq 0$; else $e_i^p = 0$.

that is the case, then z_i^p must be 1. Otherwise, if $e_i^p = 0$, then no incentive constraint is needed and z_i^p is set to 0. Thus the general problem involves combinatorial choices over which worker–project pairs receive positive effort and therefore which incentive constraints must hold.

To see an explicit example of parameters where the feasible set is nonconvex, take a simple example with one worker i and two projects, p and q. Let the productivity parameters be $\beta_i^p = 4$, $\beta_i^q = 1$, $c_i = 0.5$, $r_i = 0.3$. He prefers working on project p four times as much as on project q. The feasible set contains all pair of nonnegative efforts (e_i^p, e_i^q) that satisfy:

$$\begin{aligned} 4 &\geq 0.5e_i^p + 0.3e_i^q \qquad \text{if } e_i^p > 0 \\ 1 &\geq 0.5e_i^q + 0.3e_i^p \qquad \text{if } e_i^q > 0 \end{aligned}$$

Figure 9 illustrates the boundary of the set of implementable efforts. Effort vectors of the form $(e_i^p, 0)$ for $e_i^p \in [3\frac{1}{3}, 8]$ are feasible, as $0.5e_i^p \leq 2$. Effort e_i^q is zero at those efforts, and so his incentive constraint for project q has no bite.

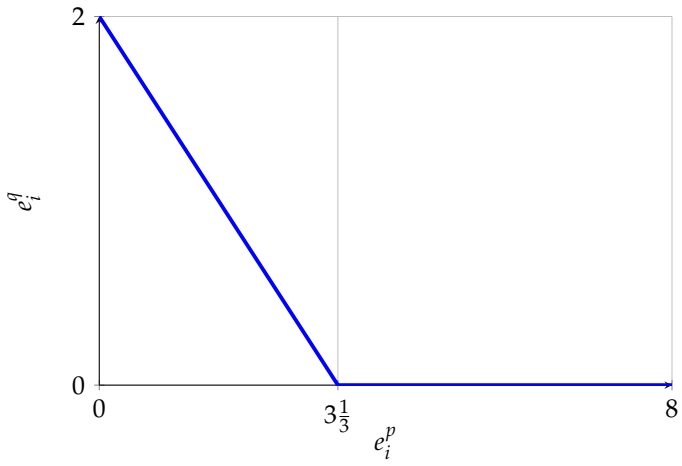


Figure 9: Nonconvex Boundary of Implementable Efforts

The inherent difficulty in analyzing the MINLP comes from the combinatorial nature of the problem. Traditional approaches such as using a greedy algorithm fail because it is neither monotone nor submodular. Increasing the effort cap for a worker at a particular project may cause him to reduce efforts at other projects, leading to an overall decrease in the manager's valuation. Raising effort caps could also cause other workers to increase their efforts on that particular project due to complementarity,

increasing the manager's valuation in a convex fashion.

Algorithmic methods, such as outer approximations, serve as efficient ways to compute the solution to the MINLP. Unfortunately, algorithmic approaches provide little in terms of structural and economic insights. Alternatively, absent the sufficient condition on the model parameters in Lemma 3, we can interpret the comparative statics exercises as local analysis treating the binary $\mathbf{z}$ variables in the MINLP as fixed.

C Generic Strict Complementarity in the Optimum

I start by forming the dual in a form that we can apply results from Drusvyatskiy and Lewis (2011). Let $V^* := \max_{\mathbf{e}\geq\mathbf{0}} V(Q^1,...,Q^m)$ s.t. $\mathbf{We} \leq \boldsymbol{\beta}$ be the primal solution where $\mathbf{W}$ is the worker interaction matrix defined in Definition 5. Forming the Lagrangian

$$\mathcal{L}(\mathbf{e},\boldsymbol{\lambda}) = V(Q^1,...,Q^m) + \boldsymbol{\lambda}^\intercal(\boldsymbol{\beta} - \mathbf{We})$$

where $\mathbf{e},\boldsymbol{\lambda} \geq \mathbf{0}$. Note that $V^* \leq \sup_{\mathbf{e}\geq\mathbf{0}} \mathcal{L}(\mathbf{e},\boldsymbol{\lambda})$ for every $\boldsymbol{\lambda} \geq \mathbf{0}$. Rearranging:

$$\sup_{\mathbf{e}\geq\mathbf{0}} \mathcal{L}(\mathbf{e},\boldsymbol{\lambda}) = \boldsymbol{\beta}^\intercal\boldsymbol{\lambda} + \sup_{\mathbf{e}\geq\mathbf{0}} \Big(V(Q^1,...,Q^m) - \boldsymbol{\lambda}^\intercal\mathbf{We} \Big)$$

Define $h(\mathbf{y}) := \sup_{\mathbf{0}\geq\mathbf{0}}\{V(Q^1,...,Q^m) - \mathbf{y}^\intercal\mathbf{e}\}$ and since $\boldsymbol{\lambda}^\intercal\mathbf{We} = (\mathbf{W}\boldsymbol{\lambda})^\intercal\mathbf{e}$ by symmetry of $\mathbf{W}$, we have that $\sup_{\mathbf{e}\geq\mathbf{0}} \mathcal{L}(\mathbf{e},\boldsymbol{\lambda}) = \boldsymbol{\beta}^\intercal\boldsymbol{\lambda} + h(\mathbf{W}\boldsymbol{\lambda})$. Now the dual problem is given by

$$\min_{\boldsymbol{\lambda}\geq\mathbf{0}} \boldsymbol{\beta}^\intercal\boldsymbol{\lambda} + h(\mathbf{W}\boldsymbol{\lambda})$$

Next, define $f(\boldsymbol{\lambda}) := h(\mathbf{W}\boldsymbol{\lambda}) + \delta(\boldsymbol{\lambda} \mid \mathbb{R}^{mn}_+)$, where $\delta(\cdot \mid C)$ is the indicator function of a convex set C in the sense of Rockafellar (1970) which equals 0 on C and $+\infty$ off C, so the dual problem can be written as

$$\min_{\boldsymbol{\lambda}} f(\boldsymbol{\lambda}) + \beta^\intercal\boldsymbol{\lambda}$$

Here f is proper and convex and does not depend on $\boldsymbol{\beta}$, so the dual is a linear tilt of a fixed convex function.[19] Slater's condition holds at $e = 0$, so there is no duality gap and the dual solution is given by $\boldsymbol{\lambda}^*$. It is unique because the binding constraints are linearly independent.

[19] f is finite at $t(\mathbf{P}^{-1}\mathbf{1}\otimes\mathbf{1}_m)$ for large t, since $\mathbf{W}(\mathbf{P}^{-1}\mathbf{1}\otimes\mathbf{1}_m) = \mathbf{1}$ and V is concave.

By Corollary 3.5 of Drusvyatskiy and Lewis (2011), for almost every $\boldsymbol{\beta}$ the dual solution is nondegenerate: $-\boldsymbol{\beta}$ lies in the relative interior of the subdifferential $\partial f(\boldsymbol{\lambda}^*)$. Let $\mathcal{M}$ be the set of maximizers of the Lagrangian at $\boldsymbol{\lambda}^*$. Standard results from Rockafellar (1970) gives $\partial f(\boldsymbol{\lambda}^*) = -\mathbf{W}\mathcal{M} + N_{\mathbb{R}^{mn}_+}(\boldsymbol{\lambda}^*)$, where $N_{\mathbb{R}^{mn}_+}(\bar{\lambda}) = \{\mathbf{z} \in \mathbb{R}^{mn} \mid z_k = 0 \text{ if } \lambda^*_k > 0;\ z_k \leq 0 \text{ if } \lambda^*_k = 0\}$. Nondegeneracy reads

$$\boldsymbol{\beta} = \mathbf{We} + \mathbf{s}, \qquad \mathbf{e} \in \mathcal{M}, \qquad s_k > 0 \iff \lambda^*_k = 0.$$

Strict concavity of V in Q gives $\mathcal{M} = \{\mathbf{e} \geq \mathbf{0} \mid \boldsymbol{\mu}^\top \mathbf{e} = \mathbf{0}\}$. This is a product of simplices scaled by $Q^{p*} > 0$, so points of $\mathcal{M}$ satisfy $e_k > 0$ whenever $\mu_k = 0$. Since $\mathbf{We} \leq \boldsymbol{\beta}$, the profile $\mathbf{e}$ is optimal, with multipliers $(\boldsymbol{\lambda}^*, \boldsymbol{\mu})$. At $\mathbf{e}$, we have (1) a binding incentive constraint has $s_k = 0$, hence $\lambda^*_k > 0$ and (2) zero effort has $\mu_k > 0$. □

D An Endogenous Sorting Index

Why is it that once the manager leaves workers off some projects in the optimum, the ordering of the cutoffs cannot be read off the primitives? Endogeneity resides in the numerator of the scope index. In general, it is worker i's total shadow value $\Lambda_i = \sum_{p=1}^m \lambda_i^p$ where λ_i^p is the multiplier on worker i's incentive constraint at project p. It measures what the manager gains if she could relax worker i's incentive constraints at every project. Under full assignment, the total shadow value is proportional to a worker i's Katz–Bonacich centrality in $\mathbf{P}$. However, this is not true when a worker is assigned to fewer projects in the optimum. His total shadow value depends on his marginal cost of total effort, which is no longer $c_i + r_i(m-1)$. Instead, it depends on the number of projects he is assigned to, which is endogenous. Thus, Λ_i is not related to $\mathbf{P}$ and depends on the assignment pattern itself.

In general, assignment and cap cutoffs are ordered by an endogenous index I call the *congestion index*. It is formulated recursively: once a worker is dropped from a team, he stops generating complementarities at that project and the network on which the index is defined shrinks as we descend the project ranking.

Definition 11. For a subset of workers $S \subseteq N$, the congestion index of worker j on $\mathbf{S}$ is:

$$\eta_j(S) := \frac{((\mathbf{M}^S)^{-1}\mathbf{R}^S\boldsymbol{\Lambda}^S)_j}{((\mathbf{M}^S)^{-1}\mathbf{1}^S)_j}$$

where $\mathbf{M}^S$, $\mathbf{R}^S$ $\mathbf{\Lambda}^S$ and $\mathbf{1}^S$ are the submatrices and subvectors with indices restricted to those in S and $\mathbf{\Lambda} := [\sum_{p=1}^m \lambda_1^p \cdots \sum_{p=1}^m \lambda_n^p]^\intercal$ where λ_i^p is the multiplier on worker i's incentive constraint at project p in the optimum. The ordering of workers is recursive. Set $N_0 := N$ and for $k = 1, ..., n$, compute

$$i(k) := \arg\max_{j \in N_{k-1}} \eta_j(N_{k-1}) \text{ where } N_k := N_{k-1} \setminus \{i(k)\}$$

Relabel worker indices in order $i(1), ..., i(n)$. Worker i's **congestion index** is given by $\eta_i := \eta_i(N_{i-1})$.

The congestion index η_i is an endogenous ratio of two centralities which captures the trade-off. The denominator is the canonical Katz–Bonacich (KB) centrality $((\mathbf{M}^S)^{-1}\mathbf{1})_i$ with the team interaction matrix restricted to members in S. It counts the total additional effort that comes from relaxing worker i's incentive constraint at a project, incorporating both the direct effect for worker i and all of the complementarity effects for other workers. The numerator applies the same team interaction matrix to Λ_i, which is the total shadow value of worker i's incentive constraints. It measures the opportunity cost incurred by the manager for all other projects when she increases worker i's effort towards a specific project, scaled by r. A high congestion index means it is better to concentrate that worker on only a few projects.

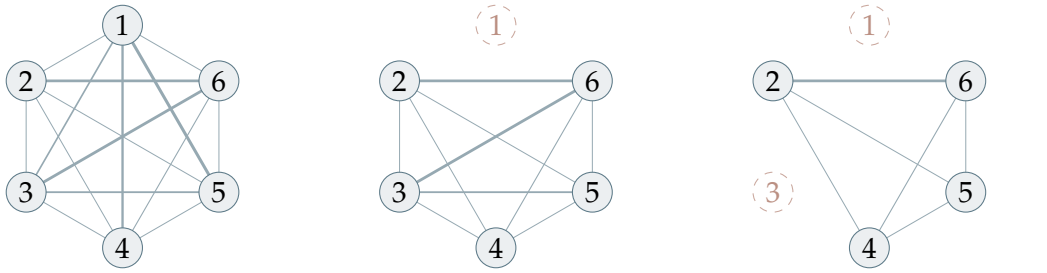


Figure 10: Once a worker is removed from a project, he stops generating complementarity to those who remain, so $\eta_i(\cdot)$ is recomputed on a strictly smaller network.

Once a specialist is identified and left off projects, he is no longer a source of complementarity for any worker left at those projects. The congestion index is recomputed for the remaining workers on the submatrix of $\mathbf{M}$ *leaving out* worker i. This result is descriptive rather than prescriptive and I emphasize that sorting cannot be achieved through just the primitives because the multipliers $\mathbf{\Lambda}$ cannot be solved in closed form.

Theorem 5. *The assignment and cap cutoffs α_i and ϕ_i are sorted such that $\eta_i > \eta_j$ only if $\alpha_i \leq \alpha_j$ and $\phi_i \leq \phi_j$.*

Proof. The linear complementarity problem that the multipliers solve is given by

$$\boldsymbol{\mu}^p = \mathbf{M}\boldsymbol{\lambda}^p + \mathbf{R}\boldsymbol{\Lambda} - V^p\mathbf{1}_n \; ; \; \boldsymbol{\lambda}^p, \boldsymbol{\mu}^p \geq \mathbf{0} \; ; \; (\boldsymbol{\lambda}^p)^\intercal \boldsymbol{\mu}^p = \mathbf{0}$$

To understand when workers are unassigned from projects due to a decreasing marginal value of projects V^p, which is a discrete parameter, I turn to its continuous counterpart.

$$\boldsymbol{\mu}^p = \mathbf{M}\boldsymbol{\lambda}^p + \mathbf{R}\boldsymbol{\Lambda} - \eta\mathbf{1}_n \; ; \; \boldsymbol{\lambda}^p, \boldsymbol{\mu}^p \geq \mathbf{0} \; ; \; (\boldsymbol{\lambda}^p)^\intercal \boldsymbol{\mu}^p = \mathbf{0}$$

V^p is replaced by $\eta \in \mathbb{R}$. I employ a pivot-type method. Ignoring non-negativity of $\boldsymbol{\lambda}^p$, I solve the system assuming that $\boldsymbol{\mu}^p = \mathbf{0}$, i.e., all workers are active on project p. We obtain:

$$\mathbf{M}\boldsymbol{\lambda}^p + \mathbf{R}\boldsymbol{\Lambda} - \eta\mathbf{1}_n = \mathbf{0} \implies \boldsymbol{\lambda}^p = \mathbf{M}^{-1}(\eta\mathbf{1}_n - \mathbf{R}\boldsymbol{\Lambda})$$

If η is sufficiently large, then $\boldsymbol{\lambda}^p > \mathbf{0}$ and all workers are assigned positive effort on project p. As η decreases, $\boldsymbol{\lambda}^p$ likewise decreases. The first index i of $\boldsymbol{\lambda}^p$ to reach zero is when

$$\eta = \left(\frac{(\mathbf{M}^{-1}\mathbf{R}\boldsymbol{\Lambda})_i}{(\mathbf{M}^{-1}\mathbf{1})_i} \right)$$

and it becomes negative if

$$\eta < \left(\frac{(\mathbf{M}^{-1}\mathbf{R}\boldsymbol{\Lambda})_i}{(\mathbf{M}^{-1}\mathbf{1})_i} \right)$$

Thus the index i corresponding to $\arg\max_i \left(\frac{(\mathbf{M}^{-1}\mathbf{R}\boldsymbol{\Lambda})_i}{(\mathbf{M}^{-1}\mathbf{1})_i} \right)$ is the first worker to be assigned $\mu_i^p > 0$, else $\lambda_i^p < 0$ which contradicts the latter's nonnegativity. This implies worker i must be unassigned from all projects p with

$$V^p < \left(\frac{(\mathbf{M}^{-1}\mathbf{R}\boldsymbol{\Lambda})_i}{(\mathbf{M}^{-1}\mathbf{1})_i} \right)$$

On those projects, we can employ the same strategy, but now restrict attention to the first index j of $\boldsymbol{\lambda}^{p,N\setminus\{i\}}$ to reach zero, which is when

$$\eta = \left(\frac{((\mathbf{M}^{N\setminus\{i\}})^{-1}\mathbf{R}^{N\setminus\{i\}}\boldsymbol{\Lambda}^{N\setminus\{i\}})_i}{((\mathbf{M}^{N\setminus\{i\}})^{-1}\mathbf{1}^{N\setminus\{i\}})_j} \right)$$

This is precisely because worker i is no longer an active participant on those projects. The same logic extends recursively to all remaining workers. □